

REVIEW

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Impact of biochar amendment on improving nutrient availability and stress resilience for sustainable crop production

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Abstract

Biochar is a stable, carbon rich by product obtained from pyrolysis of agricultural biomass and its amendment in soil represents a promising ecofriendly frontier in sustainable crop production. Production of biochar using various pyrolysis techniques offers a sustainable strategy for management of agricultural residues, forestry wastes, and other organic materials. Amendment of biochar (BC) in the soil improves soil structure and nutrient retention while serving as a supportive habitat for colonization of beneficial microbial communities that enhance nutrient availability and nutrient use efficiency, resulting in reduced fertilizer dependency of crop plants. Moreover, application of biochar has also emerged as a successful approach for boosting soil carbon sequestration, immobilization of heavy metals and organic pollutants, and in mitigation of greenhouse gases, thus contributing to ecological environment protection. Biochar is currently also gaining attention as a potentially valuable input for mitigating the impact of climate change-induced abiotic and biotic stresses in agriculture. In addition, fertilization with biochar alone or combined treatment with organic matter and/or beneficial microbes represents a promising approach in sustainable crop production in nutrient deficient soils under diverse agroecological conditions. However, the effectiveness of biochar varies with feedstock type, pyrolysis conditions and soil-crop-climate interactions. Therefore, tailored approaches considering all these factors are essential to maximize the benefits from biochar amendment. This review highlights current knowledge and technological challenges by linking fundamental mechanisms with applied outcomes for translating biochar-microbial innovations into commercially viable, environmentally sustainable and climate-resilient agricultural systems.

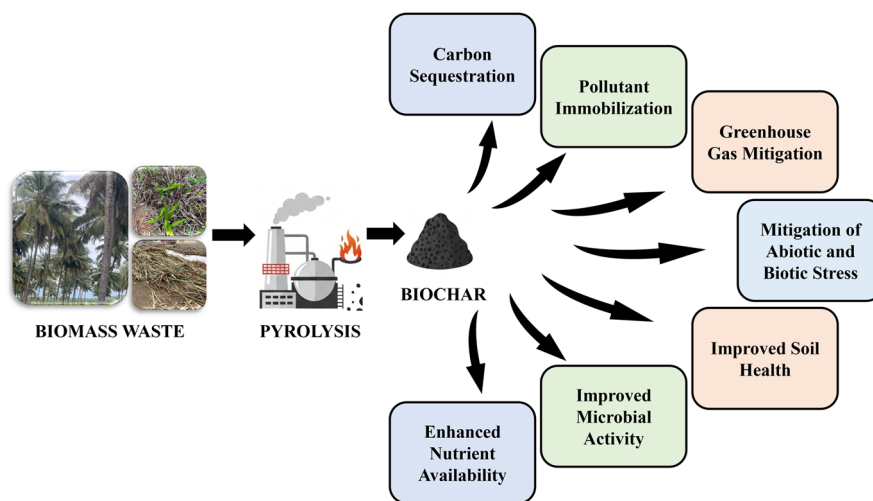
Highlights

- Biochar production is a sustainable strategy for management of agricultural residues.
- Soil fertility and nutrient use efficiency are improved by biochar application.
- Biochar alters microbial community in soil and alleviates environmental stresses.
- Biochar treatment boosts carbon sequestration and mitigates green house gases.
- Plant growth and crop yields are increased by biochar amendment.



Keywords Biochar, Pyrolysis, Sustainable waste management, Soil remediation and stress mitigation, Greenhouse gas emissions, Crop production

Graphical Abstract



1 Introduction

A major worldwide concern in today's scenario is maintaining agricultural food production as the world's population continues to grow. Under current intensive agricultural practices, safeguarding its most precious resource "soil" is a formidable challenge, since soil nutrients and organic matter (OM) are being rapidly depleted [1, 2]. The challenge is further compounded by an impending climate change issue, as evidenced by rising temperatures [3], atmospheric carbon dioxide (CO₂) levels [4], and extreme weather events like droughts [5–7]. These factors further put agricultural production systems under pressure [8–10]. Under such circumstances, modern agriculture must constantly develop and adapt novel agri-technologies in order to maximize agricultural output and resource usage efficiency, while reducing the adverse environmental consequences caused by use of chemical fertilizers and pesticides in farming [11–14].

Agricultural and environmental sustainability is also threatened by generating a huge amount of organic wastes (OWs) through the global agricultural production systems [15, 16]. Several technologies including composting, vermi-composting, anaerobic digestion, production of biochar and organic phytostimulants etc. are involved in the biodegradation and management of organic wastes [17–20]. Different indigenous and/or introduced organisms such as bacteria, fungi, algae, protozoa, nematodes and arthropods contribute in biological decomposition process of these agricultural wastes [21–23]. However, conversion of agricultural bio-waste into biochar is at the leading edge of material innovation to solve environmental challenges, impending climate change issues, and global food insecurity because of its transdisciplinary application potential [24–27]. Biochar is formed through thermo-chemical processes by pyrolyzing OWs, such as plant left-overs and crop residues (CRs) under high heat in a deoxygenated atmosphere [28–32]. This process converts organic carbon (C) in biomass into a stable form of char with a long half-life that means C is retained in a solid state [33, 34]. This helps to reduce the amount of C released into the atmosphere, contributing to mitigate global warming [30, 35]. The term "biochar" is indeed derived from Greek word. The word "bio" comes from "bios,"

meaning life, and "char" is derived from "charcoal," which is the product of the carbonization of biomass. The credit for coining the term "biochar" is generally attributed to Dr. Johannes Lehmann, a soil scientist, and his colleagues. They popularized the term in the early 2000s, when they began researching the use of charcoal in agriculture and soil improvement. While Peter Read has been involved in advocating for biochar plantations, the term itself is more closely associated with the scientific community studying its applications in agriculture and environmental sustainability [36].

The production of biochar through the pyrolysis of crop straw and CRs, and its amendment in soil has been suggested globally as an effective counter measure to increase soil organic carbon (SOC) stock, improve soil fertility, enhance ecosystem functioning and for mitigating the effects of climate change on agriculture [37–39]. High C recalcitrance against microbial decay caused by biochar amendment in croplands slowed the turnover of organic carbon [40]. It also indirectly reduces the amount of nitrogen (N) required for crop production through improved N use efficiency, which may have a negative impact on the indirect greenhouse gas (GHG) emissions from the N fertilizer industry [31, 41]. Biochar has several positive impacts on soil quality as well as fertility and it can directly add nutrients to the soil [42, 43], providing a source of essential elements for plant growth, and help adjusting in soil pH [44, 45], which is crucial for optimal plant growth. Biochar amendment enhances soil porosity [46], improves water retention and drainage, and can increase the soil's cation exchange capacity [47], which affects nutrient availability to plants. Thus, addition of biochar enhances the availability of nutrients, not only by directly supplying N, P, and K [48], but also by reducing nutrient leaching and by stabilizing soil nutrient levels [49]. Thus, biochar has gained increasing attention recently due to its potential to address environmental and agricultural issues [50, 51].

The effectiveness of biochar as a soil amendment is strongly influenced by its properties and structure. The feedstock type (the source material for biochar production) and the conditions under which biochar is produced through pyrolysis have significant impact on its characteristics. Feedstocks like woody residues generally produce biochar that is more stable, with a larger surface area and a higher porosity, while biochar made from manure or crop residues usually contains more nutrients but have lower structural stability [52]. High pyrolysis temperatures are reported to increase biochar surface area which in turn can contribute to the biochar's effectiveness in soil amendments [53–55]. The large surface area of biochar provides binding sites that can retain and exchange nutrients in the soil as the enhanced surface area can contribute to increased availability of macro-nutrients in the soil [56]. The binding sites on biochar can hold onto nutrients, preventing them from leaching away and making them more accessible to plants. High pyrolysis temperatures are associated with excellent stability of biochar in the environment, suggesting its resistance to degradation [57–59]. Moreover, higher pyrolysis temperatures are also linked to higher pH and increased mineral nutrient contents in biochar, which can have positive effects on soil properties and fertility [60]. But, biochar produced at high temperatures tends to have lower N content in some cases and may exhibit lower nutrient availability for plant uptake [61, 62]. The high porosity of biochar creates a physical environment that serves as a refuge for beneficial soil microorganisms [17]. This structure can provide a protective space away from potential disturbances caused by factors like grazing animals. Stimulation of beneficial microorganisms due to biochar amendments in field soil play a crucial role in soil health and fertility by

contributing to nutrient cycling, OM decomposition, abiotic stress resilience and disease suppression [38, 39, 63–69].

So far, several research articles and reviews have been published that discuss various approaches used for OWs management and application of these agri-technologies in crop production [15, 19, 70]. Despite the well-established benefits of biochar for improving soil health, there is a lack of comprehensive studies examining its combined effects with beneficial microbes on different biological processes including nutrient release, stress resilience, mitigation of pollution, plant growth and crop production (although see, [51, 66–68, 71, 72]), and only few findings and technologies have been translated into the field. Recently, some studies have reviewed the technology and strategies of biochar production emphasizing the qualitative properties of varying feedstocks, their effect on quantitative yield, and highlighting their environmental applications including C sequestration, greenhouse gas emissions and economic impacts [25, 72–74]. Application of biochar have also been reported to significantly enhance the nutrients in the rhizosphere and it reduced the bioavailability of heavy metals (HMs), and acted in environmental rehabilitation for remediating contaminated soil [27, 54, 55, 75, 76]. A broad overview has been provided in this manuscript about the different strategies employed for biochar production from various OWs, and the recent advancements about using biochar in enhancing nutrient release and soil fertility, in remediation of organic pollutants and in reducing GHG emissions for environmental sustainability. Furthermore, the review highlighted the importance and feasibility of modifying biochar-soil-microbiomes-plant interactions to maximize their synergies for increasing nutrient availability leading to stimulation of plant growth and increasing the crop yields for food security. In addition, this review highlights the necessity of utilizing biochar as organic fertilizer for achieving the Sustainable Development Goals (SDGs), such as zero hunger, poverty, and climate change mitigation by enhancing farm productivity. However, further in depth studies are needed to explore the implications of biochar in bioremediation of pollutants and heavy metals, and mitigation of abiotic and biotic stresses.

2 Agro-residues used for biochar production

Biochar, as a carbon-rich material produced from organic waste biomass, has gained significant attention due to its versatile applications in agriculture and potential environmental benefits [74, 75, 77, 78]. It is produced by a pyrolysis process involving the thermal decomposition of biomass (plants, food waste, and animal waste) [17, 79–81]. Lignocellulosic biomass (LB) such as rice husk, cassava stem, rhizome, sugarcane bagasse, corncob, wood, and wood chips can be used to produce biochar [50, 82–84]. LB is a widely available and renewable resource on earth and its composition varies based on factors like plant type, growth conditions, plant part, and age at harvesting [85–88].

2.1 Crop waste/residues

Increasing crop productivity needed to meet the increased food demand for exponentially rising global population has caused significant increases in crop residues (CRs). The annual total production of crops has surged to 4800 million tons, according to the Food and Agricultural Organization [89]. This surge inevitably contributes to a higher volume of CRs, which are the remnants left in the field after harvesting, and can be utilized for various purposes, including bio-based materials, fuels, and energy [20, 90]. However, the

conventional methods of disposing of these CRs, particularly for land preparation, have led to environmental issues including release of significant amount of pollutants in the air that in turn, can have adverse effects on human health as well as loss of nutrients through leaching or volatilization [15]. While CRs have the potential to serve various beneficial purposes such as livestock feed, plant nutrient reservoirs, and C sources for soil improvement, the prevalent practice of in situ incineration of CRs worldwide poses environmental challenges. This incineration leads to the release of various gaseous pollutants, including nitrogen oxides (NO_x), sulfur oxides (SO_x), carbon monoxide (CO), ozone, fine particles, as well as GHG like CO₂ and methane (CH₄) [18, 91]. In addition, the in situ burning of CRs has negative consequences for soil health and nutrient cycling [92]. It prevents effective nutrient recycling, results in the loss of C in the soil, and can harm soil microbial communities. To effectively address these challenges, it is crucial to develop specific processes tailored to the characteristics of each type of CRs. The variation in generation quantity, as well as the chemical and physical properties of different CRs, necessitates the implementation of customized approaches for their utilization and disposal. This highlights the need for sustainable and environmentally friendly alternatives to the traditional disposal methods of CRs, taking into account the potential benefits these residues can offer to agriculture and soil health [16, 70]. Efforts to find innovative ways to manage CRs, such as exploring bio-based materials, energy production, or other sustainable utilization methods, can help address the environmental challenges associated with their disposal and contribute to more sustainable agricultural practices [15, 21, 75]. It's important to recognize that the types and quantities of CRs generated can vary significantly from one location or country to another, as shown in Table 1. The major CRs utilized for production of compost, biochar, fuels and energy are enlisted in this section.

2.1.1 Sugarcane bagasse

Sugarcane bagasse is a fiber waste generated during the extraction of sugarcane. It is produced at a rate of approximately 250–280 kg per ton of processed sugarcane. This results in a substantial annual production of nearly 500 million tons of sugarcane bagasse globally. Brazil is a major contributor to sugarcane cultivation, accounting for approximately 41% of the total [94]. Biochar was synthesized from sugarcane bagasse and corncob using a vacuum pyrolyzer, to harness its possible potential applications for sustainable agricultural and environmental solutions [95]. Biochar with porous structures was created in a slow pyrolyzer at three temperature ranges. The pyrolysis temperature had a negative effect on biochar yield and reduced from 30.21 to 25.4% and 34.59 to 24.28% as the pyrolysis temperature increased from 500 to 700 °C for corncob and sugarcane bagasse, respectively. The biochar formations were discovered to possess a higher proportion of elemental carbon (90–91%) compared to their original state (43.9–45.2%).

Table 1 Agricultural wastes generated by different countries [93]

Countries	Major crops	Waste generated
Brazil, China, Thailand	Sugarcane	Sugarcane bagasse
United States, China, Brazil,	Corn	Corn stover
China, India, United States, Russia	Wheat	Wheat straw
India, China, Pakistan	Rice	Rice straw, Husk
Malaysia, Thailand, Indonesia	Palm	Kernel shells, palm fibers

2.1.2 Corn stover

Corn stover encompasses various plant parts such as leaves, stalks, and cobs, remaining in the field after corn harvesting [96]. It is considered as a sustainable material for biorefineries due to its abundance and low cost. The generation of corn stover is estimated at 1 kg per kilogram of corn grain. This leads to a substantial global annual production of approximately 1162 million tons of corn stover. The United States is the world's largest producer of corn stover, followed by China and Brazil.

2.1.3 Wheat straw

Generally, straw output can be calculated based on the grain-to-straw ratio. The output of wheat straw is 0.86 that can be correlated with its generation. This ratio implies that nearly 650 million tons of wheat straw is generated worldwide annually. China is the leading country in wheat production, followed by India, the United States, and Russia. The combined share of wheat straw production in these four countries exceeds 45%.

2.1.4 Rice straw

The production ratios of rice straw vary from 0.7 to 1.5. On an average, the annual production of rice straw is reported to be about 730 million tons [16, 97]. China and India are the leading countries in rice production, and their combined share of world rice straw production is 50%. These figures underscore the substantial amounts of straw generated globally, particularly from major cereal crops like wheat and rice. The grain-to-straw ratio provides a useful metric for estimating straw output and understanding the potential availability of this agricultural residue. Efficient management and utilization of wheat and rice straw can contribute to sustainable agricultural practices, including the production of bio-based materials, fuels and energy [18, 85]. Additionally, understanding the leading countries in straw production highlights regions where innovative strategies for straw utilization and disposal can have a significant impact. Efficiently converting crop residues like sugarcane bagasse and corn stover into valuable products can contribute to addressing environmental concerns associated with traditional disposal methods while also providing economic and sustainable solutions, especially in countries with large agricultural outputs.

2.2 Chemical composition of crop residues

The chemical composition of CRs is a critical factor in determining its suitability for various thermochemical conversion processes. CRs mainly consist of cellulose: 28–48 wt%, hemicellulose: 19–38 wt%, lignin: 12–32 wt% and others: 2–24 wt% [33, 98, 99]. These percentages vary depending on factors such as crop species, cultivation conditions, location, and age of crop plants [94]. Proximate analysis of CRs helps to understand the basic composition of the material and is particularly important for the use of biomass for energy or biochar production [38, 39]. It typically includes the determination of moisture content, volatile matter, fixed C, and ash content [100]. For biochar production from CRs, a higher fixed C content is often desirable. This is because fixed carbon represents the C that can potentially be converted into stable biochar, contributing to the C sequestration properties of biochar [38, 39, 101, 102]. This can be valuable in optimizing biochar production processes. CRs with higher volatile matter content are more easily applicable to combustion processes [103]. This characteristic is essential for

understanding the combustibility and energy release potential during thermochemical conversion. Understanding these components and their variations allows for more targeted and efficient utilization of CRs in various processes such as biochar production and combustion. Additionally, the chemical composition analysis provides insights into the potential applications of different crop residues in the context of sustainable energy production and bio-based materials.

2.3 Biochar production from agricultural biomass waste

Lignocellulosic agricultural biomass waste (ABW) serves as an excellent source of bio-energy and it is inexpensive, sustainable, renewable, and readily available [104–106]. Additionally, the diverse applications of ABW in various industries, from mechanical recycling to biofuel production, underscore its potential as a sustainable and renewable resource [107, 108]. ABW can be utilized through mechanical recycling, fermentation, and thermochemical conversion. Biomass is primarily composed of cellulose, hemicellulose, and lignin. These components undergo different reactions such as carbonization and decomposition during pyrolysis under various conditions [86, 109]. Due to the distinct chemical structures of cellulose, hemicellulose, and lignin, the resulting biochar exhibits differences in structure based on the precursor material. Lignin, having a complex structure with benzene rings and phenolic polymers [99], undergoes decomposition in a wider temperature range of 250 °C to 500 °C and it has been observed that biomass with higher lignin content tends to result in a higher biochar yield during pyrolysis. Cellulose, with better thermal stability than hemicellulose, decomposes at temperatures between 350 and 375 °C [110], whereas hemicellulose, being a polymer with a branched structure and short side chains, easily undergoes pyrolysis at temperatures ranging from 200 to 350 °C [111]. In the past, charcoal was a commonly used fuel for heating, and it was produced through wood biomass pyrolysis. Pyrolysis is a process where organic materials, such as wood, are heated in the absence of oxygen to produce charcoal, along with other by products like tar and gases. However, the old pyrolysis technology had several drawbacks, including low energy efficiency, slow production, and high levels of air pollution [112]. There are primarily six methods used for transformation of ABW such as torrefaction, pyrolysis, combustion, hydrothermal liquefaction, gasification and thermo-chemical conversion to biofuels (Fig. 1; Table 2).

Pyrolysis techniques are effective in the production of biochar. Different forms of pyrolysis exist: (i) rapid pyrolysis, (ii) ultra-quick/flash pyrolysis, and (iii) conventional/slow pyrolysis, (iv) vacuum pyrolysis, (v) intermediate pyrolysis, (vi) hydropyrolysis, and (vii) microwave assisted pyrolysis; based on operational characteristics including heating rate, pyrolysis temperature, and residence time. The solid material is usually modified via gradual pyrolysis to reduce the amount of oil generated. The amount of gases and oil generated is maximized by quick and ultra-fast (flash) pyrolysis. Rapid pyrolysis carried out at moderate to high heating rate causes thermal breakdown of carbonaceous materials in the absence of oxygen. It is the most widely used approach in both practical application and research with bio-oil as the major product. These processes are important in the context of biomass conversion and biofuel production, with each having its advantages and applications based on the yields of desired product and process characteristics. In conventional pyrolysis, the heating rate and residence times are not specified, suggesting a relatively slower process compared to ultra-fast pyrolysis. Ultra-fast pyrolysis mainly

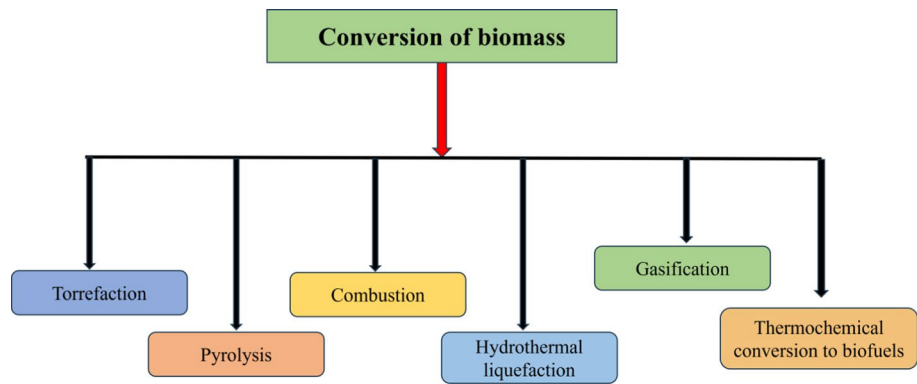


Fig. 1 Methods used for conversion of agricultural biomass waste into products. Torrefaction involves a mild thermal treatment (200–300 °C) in the absence of oxygen that produces a coal-like solid fuel and improves energy density as well as makes biomass easier to store and transport. Pyrolysis is heating of biomass without oxygen at higher temperatures. Combustion involves direct burning of biomass in the presence of oxygen. Hydrothermal liquefaction process involves high pressure and moderate temperature using water. Gasification is the partial oxidation of biomass at high temperatures and syngas is produced, whereas Thermochemical conversion involves conversion of biomass to bioethanol, biodiesel etc

Table 2 Methods of transformation of agricultural biomass waste into valuable products

Methods	Description
Thermochemical conversion	Involves using of heat to chemically transform ABW into biofuels, offering a potential solution for renewable energy production
Combustion	The combustion of biomass involves the oxidation of organic materials in the presence of oxygen to produce heat. While combustion has been a traditional and widely used method for heating and energy generation, it have certain drawbacks such as low efficiency and release of pollutants such as particulate matter, carbon monoxide (CO), nitrogen oxides (NOx), and volatile organic compounds (VOCs) [113, 114]
Hydrothermal Liquefaction	This process uses high-temperature and high-pressure water to convert ABW into bio-oil. However, it may require more pressure compared to pyrolysis
Torrefaction	Torrefaction involves roasting ABW to improve its properties for combustion. However, it may lead to tar formation
Pyrolysis	Pyrolysis is highlighted as an efficient technique to convert ABW into valuable products i.e., residues, gases and oil. It produces fewer emissions compared to combustion methods [115]
Gasification	Gasification involves converting ABW into a gas mixture, which can be used for energy production. However, it may have higher energy consumption compared to pyrolysis

produces gases and bio-oil, with a significantly higher percentage of gases compared to conventional pyrolysis [116]. The yield of liquid condensates in ultra-fast pyrolysis is lower than in conventional pyrolysis. Slow pyrolysis is used for solid material modification with minimized bio-oil production. The temperature at which biochar is produced also affects mineralization and immobilization processes, for example, biochar formed at lower temperatures (250 °C–400°C) tends to encourage mineralization, whereas high temperature biochar (525 °C–650°C) is more conducive to long-term carbon storage and these variations were due to the structural changes caused by pyrolysis processes [117].

2.3.1 Conventional/slow pyrolysis

Slow pyrolysis is characterized by a long residence time, exceeding one hour. During this process, biomass is heated at temperatures ranging between 300 and 600 °C, with a gradual heating rate of 5–7 °C/min [118–120]. The slow heating rate and extended residence time create an environment conducive to secondary reactions. Secondary reactions refer

to chemical transformations that occur after the initial pyrolysis step, contributing to the development of biochar. The prolonged residence period facilitates the evacuation of vapours generated during these secondary reactions. In this process, the primary product biochar accounts for 35–45% of the overall products [121], bio-oil constitutes 25–35% of the products, and syngas accounts for 20–30% of the by-products [122, 123].

2.3.2 Rapid/fast pyrolysis

Fast pyrolysis is a direct thermochemical process that rapidly converts solid biomass into liquid bio-oil. It is conducted in the absence of O₂, at temperatures above 500 °C, and with a high heating rate of over 300 °C/min. Fast pyrolysis is designed to take biomass to temperatures where thermal cracking occurs rapidly, minimizing exposure time [124]. This approach supports efficient biochar production while emphasizing the liquid bio-oil as a primary output [125, 126]. The process typically yields 60–70% bio-oil, 15–20% biochar, and 10–20% syngas [127–129]. This distribution makes it a versatile method for obtaining liquid biofuels, solid biochar for soil improvement and syngas for potential energy applications.

2.3.3 Flash/ultra-quick pyrolysis

Flash pyrolysis is presented as an enhanced and modified version of fast pyrolysis, characterized by extremely rapid biomass decomposition. The process is conducted at temperatures between 900 and 1200 °C, reaching these temperatures in less than one second (typically between 0.1 and 1 s) [130, 131]. The rapid decomposition of biomass and short exposure to high temperatures result in a swift vaporization and collection of bio-oil. The rapid nature of the process may limit the formation of solid carbonaceous biochar.

2.3.4 Vacuum pyrolysis

Thermal decomposition of biomass is carried out under vacuum or relatively low pressure in an isolated O₂ environment [132, 133]. Vacuum pyrolysis is carried out under reduced pressure, typically in the range of 0.5 to 2 bar (0.05–0.2 MPa), and at controlled temperatures maintained between 450 and 600 °C [134]. The effective evacuation of vapours is made to minimize residence time and prevent secondary reactions.

2.3.5 Intermediate pyrolysis

Intermediate pyrolysis integrates aspects of both slow- and fast- pyrolysis processes. This combination is chosen when there is a need to balance the production of solid (char) and liquid products. Slow pyrolysis is known for efficiently producing large amounts of char but results in lower amounts of liquid products. Conversely, fast pyrolysis is more effective at producing liquid products but generates less char. Thus, intermediate pyrolysis aims to strike a balance between these two extremes. The process is typically conducted at a pressure of 1 bar, with temperatures ranging between 500 and 650 °C. Heating rates fall between 0.1 and 10 °C/min, and the residence time is in the range of 5 to 17 min [135]. Intermediate pyrolysis conditions are designed to prevent the synthesis of high molecular reactive tars. This contributes to the production of dry biochar suitable for agricultural purposes or for direct use in boilers and engines [136].

2.3.6 *Hydropyrolysis*

Hydropyrolysis involves the conversion of biomass by injecting hydrogen or hydrogen-based materials into a reactor under high pressure. The pressure typically ranges from 50 to 200 bar [133, 137]. The operational conditions for hydropyrolysis include a heating rate of 10–300 °C/s, a residence time of over 15 s, and a temperature ranging from 350 to 600 °C [138]. These conditions are comparable to those of fast pyrolysis, making hydropyrolysis a special type of fast pyrolysis under high-pressure conditions with the introduction of hydrogen. This process reduces the O₂ content in the bio-oils produced, simultaneously inhibiting the production of biochar. Therefore, hydropyrolysis is not considered ideal for biochar production [139, 140].

2.3.7 *Microwave-assisted pyrolysis (MAP)*

This is the most appropriate pyrolysis technique available because it uses volumetric heating to convert ABW into biochar [141]. This technique involves the use of microwaves to uniformly heat the entire volume of a material, whether it's a solid, suspension, or liquid. Unlike conventional heating methods that rely on radiation, conduction, and convection, volumetric microwave heating converts microwave energy into thermal energy throughout the material's volume [142–144]. During microwave assisted pyrolysis, materials are heated using electromagnetic waves rather than hot surfaces [145]. This approach minimizes thermal damage to the treated materials. Additionally, it allows for the heating of larger volumes of viscous materials without affecting the product flow rate through the microwave heating chamber, reducing fouling issues [146]. In microwave heating, the higher the temperature of the final product, the slower the flow [147]. Overall, volumetric microwave heating presents a modern and efficient alternative to traditional thermal processing methods, offering benefits such as uniform energy distribution, reduced thermal damage, and the ability to handle viscous materials effectively on an industrial scale.

2.4 *Production of functional biochar for agricultural sustainability*

The production and designing of functional biochar from agricultural biomass wastes is a promising avenue with potential benefits for both agriculture and environmental sustainability [148–150]. The biochar production also helps in reducing reliance on chemical fertilizers and controls the in situ burning of crop waste as shown in Fig. 2 [151]. For designing functional biochar from wastes, Han et al. [152] explored the effects of FeCl₃ and AlCl₃ addition to two typical biomass wastes (rice straw and poultry litter) during pyrochar and hydrochar production. The addition of Al (III) facilitated the carbonization of 250 °C pyrochar, but suppressed that of hydrochar and 450 °C pyrochar. The addition of Fe (III) enhanced the formation of aromatic structures of both pyrochar and hydrochar, but reduced the abundance of lignin- and condensed aromatics-like species in their dissolved organic C fractions. Furthermore, Fe (III) increased the stability and adsorption efficiency of pyrochar and hydrochar for both polar and nonpolar contaminants. Thus, functional biochar may be designed from biomass wastes with high stability and adsorption efficiency for organic contaminants.

In another experiment, the effect of pyrolysis temperature was evaluated systematically on the structure of biochar prepared from crop straw obtained from three crops (cotton, alfalfa, and wheat) [76]. It was observed that higher pyrolysis temperature

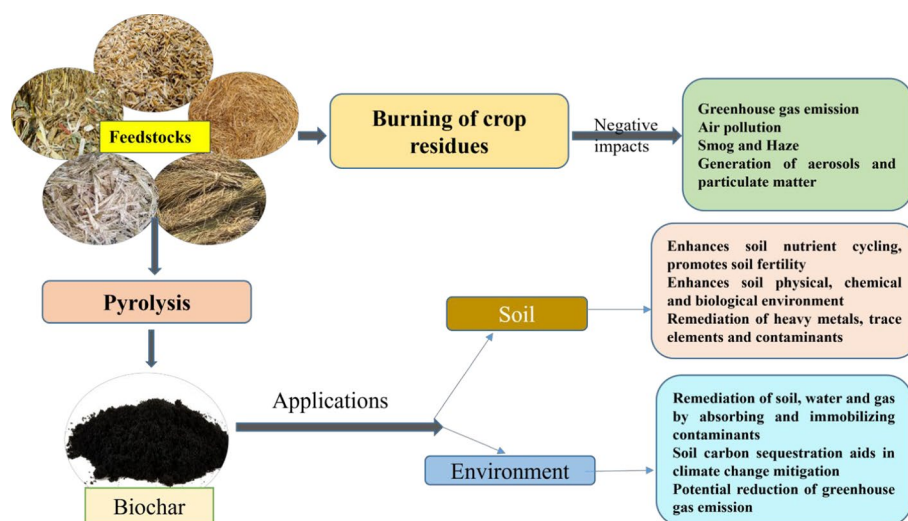


Fig. 2 Biochar production from crop residues and its benefits in agriculture and environment. There are mainly two possible pathways for management of crop residues: Traditional practice involves burning of crop residues and other sustainable alternative is its conversion to valuable products (valorization). Burning of crop residues in the traditional practice has several negative impacts that include greenhouse gas emission, air pollution and release of aerosols and particulate matter. In sustainable alternative crop residues are converted to some useful products. One such alternative includes pyrolysis of waste to produce biochar, which is carbon-rich material and its amendment in soil improves nutrient cycling, increases soil fertility, helps in mitigation of climate change by reducing emission of greenhouse gases

increased aromatization and graphitization degree of BC, and established a relationship of pyrolysis temperature with surface properties of biochar such as functional groups and carbonization degree. In addition, the content, composition, and functional group evolution of BC-derived dissolved organic matter was controlled by the pyrolysis temperature. However, pyrolysis concentrated the amount of potentially toxic elements (PTEs) in BC, but the potential risk of PTEs significantly decreased in biochar. The BC prepared at 500 °C showed the highest spin concentration of persistent free radicals. In a recent study, Malik and Laura [153] prepared biochar using agro-residues (rice husk-RH, cotton stalk-CS, orange peel-OP and sugarcane bagasse-SB) and assessed the physico-chemical characteristics as well as functional groups of biochar synthesized at 450 °C for 2 h. Elemental analysis revealed that pyrolysis led to increased C content and reduced O₂ levels across the samples. The highest C concentration was observed in CS biochar (66.30%), while SB biochar showed the lowest C concentration (19.39%). Moreover, CS biochar exhibited the highest hydrogen-to-carbon (H/C) and oxygen-to-carbon ratio (O/C) molar ratios indicating greater aromaticity and thermal stability. These characteristics make CS biochar suitable for soil improvement and carbon sequestration.

Recently, a sustainable and cost-effective technology was developed to prepare biochar-based slow-release fertilizer (BSRF) through the co-pyrolysis of wheat straw (WS), bentonite and nutrient solution containing KH₂PO₄ and KNO₃ [154]. The resulting materials exhibited biochar yields of 44.6–56.3%, pH values between 7.6 and 8.6, and C:N ratios of 7.9–22.7. Soil incubation study at 35 days demonstrated that the respective release of NO₃-N, PO₄³⁻, and K⁺ ranged from 43.2 to 59.4%, 22.7 to 66.8%, and 52.1 to 69.5%. The BSRF prepared with 30% bentonite at 350 °C pyrolysis demonstrated the best slow-release behaviour in soil. In another study, effects of pyrolysis temperature and the chemical characteristics of lignocellulosic biomass were evaluated on the release of

hazardous volatile organic compounds (HVOCs) from biochar [155]. The lowest release of HVOCs was associated with temperatures 350–425 °C and elevated carbon-relative molar mass (CRMM) index. Results provided a new approach to consider both process and substrate parameters for evaluating the safety of biochar in the context of environmental and human health protection.

3 Effect of biochar amendments on soil health

Soil pollution is identified as one of the main environmental problems worldwide. The primary contributors to soil contamination are identified as chemical fertilizers, pesticides, mineral oil and heavy metals (HMs), which are contributing around 60% to soil contamination in the European Union [156]. Application of organic amendments, particularly biochar derived from the transformation of biomass residues through pyrolysis is a valuable alternative for restoring degraded soils [157, 158]. The rehabilitation strategy involving organic amendments is noted to enhance various soil properties, including both physical and chemical properties, nutritional value, water holding capacity (WHC) and other essential parameters of soil that indirectly influence the sorption and complexation of metals in contaminated soils [75].

Many tropical and subtropical soils face severe fertility depletion due to excess nutrient mining and reduction in soil organic matter (SOM). The application of biochar is considered as an innovative solution to address the complex issues of soil quality and agricultural productivity [159, 160]. Biochar addition is noted for its ability to modulate soil biology, chemistry, and physical properties, and also to work efficiently in acidic and deteriorated coarse-textured soils. The effectiveness of biochar application may vary depending on the specific characteristics of the soil. The optimum application is suggested to have the potential to improve soil conditions, reduce GHG emissions, and immobilize toxic metals and other organic pollutants [161]. Despite its potential benefits, some adverse effects of biochar have been recognized at higher doses [162]. Studies indicate that there is a specific threshold beyond which additional biochar application becomes detrimental on plant growth or other parameters [163]. Furtado et al. [164] conducted studies on sunflower (*Helianthus annuus* L.) and observed a reduction in plant height when the rate of poultry biochar exceeded 5%. Similarly, Rajkovich et al. [165] found that the growth of corn was negatively impacted when the rate of food waste biochar was increased beyond 7%. In both cases, an increase in the rate of biochar application led to a reduction in plant growth, indicating that there might be an optimal or safe range for biochar application that should not be exceeded. Thus, the rate of biochar application is crucial, and exceeding certain thresholds may have adverse effects on plant growth [166].

The recommended biochar application rates vary greatly depending on soil type, crop system and biochar characteristics. Applying biochar at a rate of 8 tons per hectare gives an internal rate of return of 85.71% and a benefit–cost ratio of 1.476 [167]. This biochar application rate provides a balance between cost and agronomic benefits such as improved soil health and crop yield and reduced need for chemical fertilizers. Most studies report beneficial effects of biochar in the range of 5–20 t ha⁻¹, however, rates exceeding 20 t ha⁻¹ are not economically feasible and may result in negative effects such as nutrient imbalance, increased soil alkalinity and reduced availability of micronutrients

[167, 168]. This highlights the importance of considering the appropriate dosage and application methods to avoid negative consequences.

3.1 Effect of biochar on physical and chemical properties of soil

The porous structure of biochar, its large surface area, water holding capacity and content of nutrients improves physico-chemical properties and biological health of soil (Fig. 3). Biochar generally reduces bulk density in soils. This is beneficial because lower bulk density indicates increased soil porosity, which refers to the volume of pore space in the soil. This increased porosity improves water infiltration and retention reducing the risk of drought stress for plants, promoting better drainage and reducing the risk of water logging. The reduction in bulk density and alteration of soil porosity by biochar contributes to increased plant productivity [169]. The porous structure of biochar helps create a more open soil matrix, reducing soil compaction, which can impede root growth and nutrient uptake by plants. Biochar can serve as a habitat for beneficial microorganisms due to its porous structure that can positively influence soil microbial activity, leading to improved soil health and nutrient cycling. Biochar can improve soil hydraulic conductivity, facilitating better water movement through the soil profile that assist in preventing surface runoff and erosion. Biochar can influence soil colour due to its inherent dark color. This effect is generally more pronounced in lighter-colored soils.

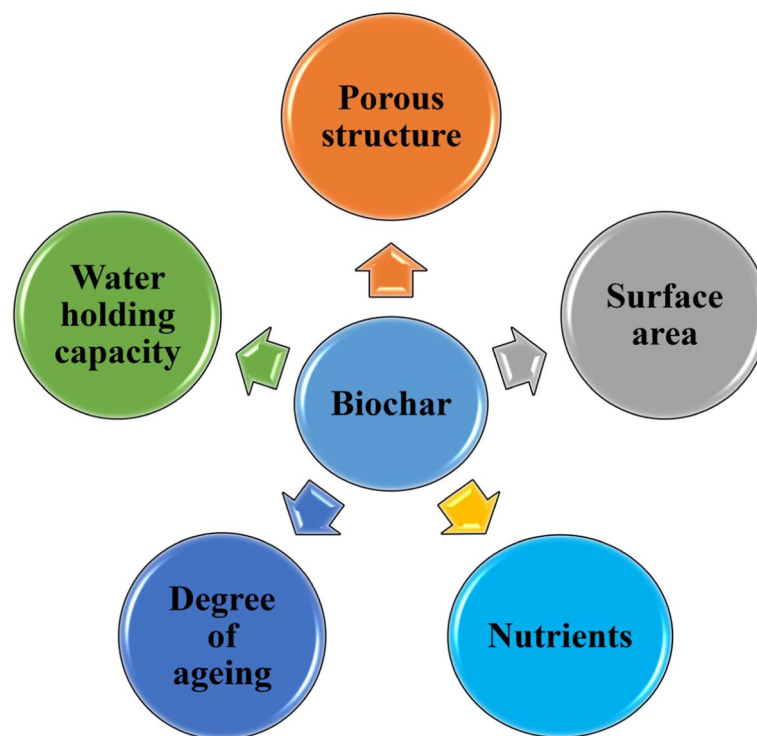


Fig. 3 Factors affecting colonization of microorganisms in biochar. The figure explains several factors that affect the colonization of microorganisms in biochar. Each factor plays a significant role in determining how effectively microorganisms can attach, grow and thrive in biochar. Due to porous structure, biochar offers protected habitats for microorganisms, shield them from predation and environmental stress. Larger surface area of biochar increases the number of sites available for microbial attachment. Biochar contains various nutrients, which serves as a nutrient reservoir for microorganisms. Biochar surface undergoes physical, chemical and biological weathering (during ageing) that increases the hydrophilicity and nutrient retention in biochar. Furthermore, biochar's ability to retain water provides a moist microenvironment for microbes that provides the microorganisms the adequate moisture and dissolved nutrients even during dry conditions

The dark color of biochar can enhance soil warming by absorbing and retaining solar radiation, which may be beneficial for certain crops and in cooler climates.

The effects of biochar on soil physical properties can vary depending on the type of biochar, feedstock, application rate, and specific soil and climate conditions [162, 170–172]. Biochar can alter the size and distribution of soil pores, leading to a shift towards relatively smaller pore sizes, which is beneficial for crop growth and development. The impact of biochar on hydraulic conductivity is noted to be more significant in sandy soils compared to clay soils [160, 173]. For instance, sandy soils experience more significant changes in hydraulic conductivity compared to clay soils [174]. In sandy soil, biochar application significantly reduces saturated hydraulic conductivity due to enhanced water retention in a single layer that contributes to better water availability for plants in sandy soils. The type of feedstock used for biochar preparation can influence water holding capacity. Woody feedstocks generally have higher WHC than lignocellulosic and herbaceous feedstocks, attributed to greater pore space.

Garbuz et al. [175] reported that adding biochar (1% w/w) made from willow wood produced at low 350 °C temperature to the two contrasting soils i.e., a sil-andic Andosol and a dystric Cambisol pasture soils positively affected the soil structure, decreased soil bulk density, improved nutrient cycling (soil total N and Olsen P content), enhanced microbial biomass C, and increased root biomass in the soil. In a study carried out by Oyeyiola and Ogunlaran [176], the properties of biochar derived from sawdust and *Tithonia diversifolia* weed at different pyrolysis temperatures (350 and 450 °C) and with or without the addition of poultry manure were compared. The study focused on liming efficiencies, macronutrient enrichment, and C sequestration potentials in acid soil. Biochar derived from *Tithonia diversifolia* weed had higher pH, ash, basic cations, N and P contents, whereas sawdust-derived biochars had higher organic C content. The inclusion of poultry manure in the feedstock improved liming efficiency and increased macronutrient enrichment potentials for all biochars. The performance was notably better at a pyrolysis temperature of 450 °C. The study compared the properties of biochar with those of conventional CaCO₃ (calcium carbonate) in acid soil. All the biochars were found to have comparatively high liming efficiencies as that of conventional CaCO₃ but outperformed CaCO₃ in their macronutrient enrichment, C sequestration and cowpea seedling root parameters enhancement. Soil chemical properties are also crucial indicators of sustainability in crop production [177]. The nutrient status of the soil and the crop acquisition pattern are dependent on these properties. The period following the green revolution has resulted in multi-nutrient deficiencies in the soil due to the continuous overuse of fertilizers [178]. Biochar serves as an innovative nutrient management option to enhance soil nutrient effectiveness and efficiency to benefit plants by providing a continuous flow of nutrients. In a study conducted by Burrell et al. [160], woodchip derived biomass resulted in greater reductions in bulk density in coarse-textured Planosol compared to other soils. It also significantly improved soil aggregate stability in the Planosol as compared to Cambisol and no noticeable impact in Chernozem. They also reported that straw derived biochar was more effective in enhancing soil structure, particularly in the Planosol, where it increased aggregate stability by upto 98% and plant available water by 38%.

Addition of biochar to the soil also improves the overall organic C content. Organic matter is known to influence cation exchange capacity (CEC) by providing binding sites

for cations. This, in turn, improves the availability of essential minerals such as potassium (K), calcium (Ca), magnesium (Mg), zinc (Zn), manganese (Mn), and copper (Cu). Over a long time (more than 5 years), biochar additions have been shown to enhance SOM by 46%, which is crucial for the mineralization process of nutrients [170–172]. In another study by Mohan et al. [179], corn stover and rice husks were converted to slow pyrolysis biochars (CSBC and RHBC), and these biochars were then used in soil incubation studies. The pH of water suspensions containing CSBC and RHBC was higher (more basic) compared to a water suspension of control soil. The elevated pH values in biochar suspensions were attributed to the presence of basic oxides, hydroxides, and possibly carbonates formed from ions like Na^+ , K^+ , Ca^{2+} , and Mg^{2+} during the pyrolysis process that produces biochar. Increase in electrical conductivity (EC) values as well as soil pH has also been observed with the addition of biochar to soil. The rise in soil pH could be linked to an increase in the soil's cation exchange capacity which in turn is attributed to the high surface areas and porosity of biochars [180]. These changes can have implications for soil fertility, nutrient availability, and overall soil health. The application of biochar improved the availability of K, Na, Ca, and Mg, ammonium bicarbonate-diethylene triamine pentaacetic acid (AB-DTPA) extractable Fe, and Mn in calcareous loamy sand soils, while reducing the AB-DTPA extractable Zn [181].

3.2 Effect of biochar to enhance C sequestration in carbon neutralization technology

Among the global carbon cycle, soil C pool is the largest terrestrial carbon pool containing 1500–2400 Gt (Giga tonnes) C, which is 2.5–3.5 times greater than that of atmospheric CO_2 (860 Gt C) and vegetation (450–650 Gt C) [182]. Soil organic matter is an important factor among terrestrial soil C pool that influences the soil fertility and health. Since a slight change in soil C storage can profoundly impact atmospheric CO_2 concentration, soil C sequestration has emerged as a crucial aspect of mitigating GHG emissions and maintaining ecological balance [170–172, 183]. With the growing emphasis on sustainable agriculture and ecological environment protection, it has become exceedingly urgent to search for efficient soil C sequestration strategies.

Application of biochar has been demonstrated to be a successful strategy for boosting soil C sequestration and altering the agricultural soil C cycle [184–186]. The dissolved solids released from biochar not only shorten the duration of C sequestration but also diminish its efficacy on improving soil C fixation. Highly aromatic and recalcitrant structure of biochar makes it resistance to microbial degradation, helping in long term carbon storage [187]. Also, biochar can physically protect organic matter by trapping it in its porous matrix and adsorbing dissolved organic carbon on its surface due to its high adsorption ability. Biochar also enhances soil aggregation, which further stabilizes organic carbon by limiting microbial access and enzymatic breakdown [117]. El-Naggar et al. [188] recorded increase in SOM content by 42–72% in sandy and 32–48% in loam soil by adding biochar produced from rice straw, umbrella tree, silver grass and crop residues. Adding biochar to soils also increases the amount of SOC [189]. It increases SOC either by microbial decomposition or by preserving existing natural SOC [190].

The persistent OM decomposers in biochar amended soils mineralized the recalcitrant components and increased the soil derived CO_2 efflux. It has been reported that biochar produced from agricultural residues can trap approximately 1.2 t C ha^{-1} annually [191], whereas biochar derived from manure captures nearly $0.6 \text{ t C ha}^{-1} \text{ yr}^{-1}$, which

might be due to the differences in C contents [192]. Majumder et al. [193] found that biochar sequester 0.7 t C ha^{-1} annually on average basis, with observed variations ranging from 0.1 to $2 \text{ t C ha}^{-1} \text{ yr}^{-1}$ depending on factors such as feedstock type and pyrolysis conditions. In another study, faster mineralization of SOC was reported with amendment of soil with Biochar500 or Biochar600 (biochars pyrolyzed at 500°C and 600°C , respectively) [38, 39] that offset C sequestration effects. A shift in microbial community structure was also induced by addition of BC in soils, which caused the dominance of Gemmatimonadetes, Bacteroidia, Alpha proteobacteria and Gamma proteobacteria. Biochar600 increased additional soil derived CO_2 carbon loss over 28 days incubation ($93 \text{ mg C kg}^{-1} \text{ soil}$). In another experiment, the addition of BC increased the relative abundance of Actinobacteria in the rhizosphere, enhanced availability of nutrients, and increased the root biomass [65]. The expression of β -1,4-glucosidase (BG) and β -N-acetylglucosaminidase (NAG) hydrolases was enhanced by 26% and 47%, respectively by expanding the extent of rhizosphere hotspots. In contrast, increased nutrient cycling was obtained in maize rhizosphere by triggering microbial growth and biomass due to addition of straw. The straw treatment enriched especially fast-growth r-strategists (Firmicutes), and a large number of enzymes including BG and NAG hydrolases were synthesized.

Meng et al. [184, 185] performed a meta-analysis of 1150 comparisons from 86 peer-reviewed publications to assess the effects of four types of biochar application during soil C cycle on carbon components in a farmland. The application of chaff biochar significantly inhibited (10.0%) soil respiration, while the application of manure biochar (47.0%), straw biochar (11.2%) and wood biochar (8.7%) showed a strong promotion effect on CO_2 emission. In addition, soil organic C, microbial biomass C and dissolved organic C all had positive responses to the application of the four biochar types, the degree and increase in their response varied greatly due to the differences in biomass raw materials. Moreover, by increasing the biochar rates applied to coarse-textured soils with low average annual rainfall and an average temperature under controlled circumstances, the relative increase in SOC was encouraged. Meanwhile, applying low temperature pyrolytic biochar ($\leq 400^\circ\text{C}$) at a lower rate ($< 25 \text{ t/ha}$) in the long-term experiment (> 3 years) is more beneficial to soil C sequestration and emission reduction. Hence, climatic conditions, agricultural management practices, and initial soil properties jointly constrained and influenced the ability of biochar to alter the soil C cycle.

Biochar prepared from cellulase-treated lignocellulose (BC-SR) was utilized as a soil amendment to improve soil C fixation and enhance C sequestration in a carbon neutralization technology [182]. BC-SR promoted a net increase of 4.39 mg C g^{-1} in soil (equivalent to fixing $16.10 \text{ mg CO}_2 \text{ g}^{-1}$), which was 28% higher than ordinary biochar after one year. The BC-SR enhanced microbial autotrophic metabolism, promoted the formation of soil macroaggregates to protect easily degradable organic carbon and increased the proportion of refractory organic matter. Thus, BC-SR exerted a stronger effect on autotrophic metabolism as compared with ordinary biochar and the newly generated OM persisted in the soil for a longer time. In another study, low-release biochar was found more stable compared to conventional biochar due to its resistance to soil microbial degradation and the autotrophic microbial agents strengthened this property. Autotrophic microbial agents and low-release biochar (AM&BC-SR) were used as a combined soil amendment to boost soil C sequestration [194]. After one year, AM&BC-SR's soil

organic carbon was 33% higher than the initial and 78% above blank, achieving a net carbon fixation of 4.1 g kg^{-1} . Moreover, their synergistic effect was manifested in the soil physical and chemical properties and soil C metabolism. AM&BC-SR significantly promoted the most widely distributed microbial carbon fixation pathway (Calvin–Benson cycle), with the abundance of key enzyme genes being 35~120% higher than blank. In soil heterotrophic metabolism, AM&BC-SR also promoted the glycolysis pathway. The synergistic functional system among biochar, autotrophic microbial agents, and soil indigenous heterotrophic microorganisms enabled AM&BC-SR to effectively increase soil organic C while keeping the rise in soil respiration at a low level. Thus, use of AM&BC-SR has emerged as novel carbon–neutral technology that can play an unprecedented role in soil improvement as a green fertilizer.

3.3 Influence of biochar on microbial community and biological processes

The use of biochar in soil has multifaceted benefits for stimulating growth of microbial communities leading to an increase in their numbers and abundance in the soil [23, 195, 196]. The expression of plant growth promoting (PGP) activities by these microbial communities enhance soil nutrient content and soil fertility, mitigate the negative impacts of environmental stresses, and promote growth of crop plants [72, 73, 197–199]. The increased availability of nutrients also plays crucial role in enhancing the diversity and abundance of soil microbial communities [200–202]. Biochar has a high adsorption capacity due to its large specific surface area that allows it to immobilize the pollutants in soil and minimize their toxic effects on microorganisms.

3.3.1 Biochar as habitat for soil microorganisms

Application of biochar improves the physicochemical and biological properties of the soil. The porous structure of biochar is highlighted as a crucial feature with significant effects on the soil biology and environment [198, 199]. Pores in biochar serve as a habitat for microorganisms in the soil. Biochar's porous structure also plays a role in protecting soil microbes from predation [203]. This protection is particularly relevant because the pore size of biochar is typically larger than most soil microorganisms. The size of biochar pores excludes large arthropods. The ability of biochar to accommodate soil microorganisms within its pores, coupled with the exclusion of larger predators, provides a microenvironment conducive for the colonization of microorganisms in the soil [63, 64, 204]. Several factors affect microbial colonization of biochar as shown in Fig. 3. The rate of microbial colonization of biochar is linked to the degree of aging of the biochar; as aged biochar is generally more favourable for microbial growth due to reduced release of contaminants. Freshly applied biochar may release endogenous contaminants, such as cadmium, which can slow down the microbial colonization in the soil [205]. The co-interaction between different microorganisms can also play a role in determining the time and rate of colonization [206].

The application of biochar to soil has diverse effects on soil physical properties, such as density and aggregate stability, as well as on key soil processes, including water retention, GHG emissions (both N_2O and methane), and the composition of nitrifying and denitrifying bacterial communities. These changes have direct and indirect consequences for soil microorganisms and can influence the overall health and functioning of the soil ecosystem. Biochar application affects the density of the soil and the stability

of aggregates [187, 207]. The reduction in soil bulk density may contribute to improved soil structure and its water holding capacity. This improvement in water retention can positively impact the availability of water for plants and soil microorganisms. The effect of biochar application can have implications for nitrogen cycling in the soil due to its influence on the composition and activity of soil nitrifying and denitrifying bacterial communities [12]. These changes will subsequently affect nutrient availability and losses in the soil and the rhizosphere. Zheng et al. [208] reported that the application of BC in paddy soils had specific effects on the total N content and the community structure of soil microorganisms. Biochar application resulted in an increase in the total N content in paddy soils. In general, bacterial communities were found to be less affected by biochar application, whereas the communities of Ascomycota and Basidiomycota were more affected. The differential impact on bacterial and fungal communities suggests that biochar application may have a selective effect on different microbial groups.

The impact of straw biochar was evaluated on soil microbiome in the wheat rhizosphere using metagenomic sequencing [56]. It was observed that straw return decreased the yields of wheat, while the straw biochar increased the wheat yields. The straw biochar return also resulted in greater rhizosphere effects from wheat, as represented by resource availability, including SOC, soil total N, available P, and available K. However, the expression of microbial metabolism genes involved in C, N, P, and K cycling, were decreased by straw biochar returning. These results revealed that straw biochar enhanced soil resource availability, but suppressed microbial metabolism genes in the wheat rhizosphere. In another experiment, cucumber plants were treated with sweet pepper straw biochar (SPSB) at 5% (w/w) concentration and high-throughput sequencing of soil microbial community structure was performed [63, 64]. Soil microbial community structure was significantly altered with this biochar concentration and BC amendment enhanced bacterial and fungal diversity. The beneficial bacterial groups especially Firmicutes, *Actinobacteria* and *Bacillus* were increased, and a decrease in *Ascomycota* and *Aspergillus* along with shifts in *Penicillium* abundance was observed. The enrichment in bacterial metabolic pathways as well as upregulation of fungal expression and replication genes was reported using functional genomic analysis.

In another recent study, the amendment of biochar along with halotolerant *Halomonas salifodinae* was examined on *Medicago sativa* grass and it significantly increased soil pH, enzyme activities, SOC, and readily oxidizable carbon [66–68]. Moreover, silicon-modified biochar significantly increased the relative abundances of beneficial taxa such as *Bacillus* and *Xanthomonadaceae* by 482.7% and 52.7%, contributing to enhanced microbial diversity and network stability. In similar study, the effect of biochar amendment on microbial metabolic quotient ($q\text{CO}_2$) was analyzed in rice paddy field and incorporation of biochar induced a lower microbial $q\text{CO}_2$ in the bulk soil, whereas little effect on $q\text{CO}_2$ was observed in the rhizosphere compared with the non-amended control [209]. The sizes of soil labile C and N pools significantly declined in the bulk soil, whereas increased in the rhizosphere under biochar. Biochar amendment at 40 t ha⁻¹ significantly changed bacterial community composition and increased their co-occurrence network complexity. In addition, the activities of β -glucosidase, phenol oxidase and peroxidase was enhanced after biochar amendment in both bulk and rhizosphere soils. Structural equation modeling suggested that biochar decreased $q\text{CO}_2$ in the bulk soil mainly by increasing soil pH, C/N ratio, microbial substrate utilization and

enzyme activities. This study suggested that soil organic carbon lability, carbon-degrading enzyme activities and changes in bacterial community composition were key drivers of $q\text{CO}_2$ in the bulk soil.

3.3.2 Effect of biochar on microbial signaling molecules

Various kinds of signaling molecules are secreted by plants and microbes into the rhizosphere, which affect the communication and interactions of microorganisms with plants during beneficial or harmful interactions [210]. These signaling molecules control the microbial growth and coordinate the behaviour and activities of microorganisms in the rhizosphere. Various signals molecules include N-acyl homoserine lactones (AHLs), strigolactones, flavonoids, diffusible signal factor (DSF), VOCs and phytohormones [211, 212]. These molecules may also function as inter-kingdom signals that elicit various effects on plant developmental processes. The biochar application has been found to affect the communication of microbial signaling molecules. For instance, the degree of hydrolysis of signaling molecule AHL is influenced by changes in soil pH, which is commonly used by Gram-negative bacteria for quorum sensing for cell-to-cell communication, allows bacteria to coordinate their behaviour based on population density. As pH increases due to application of biochar, the hydrolysis of AHL also increases, and it results in a reduction in intercellular signaling by bacteria. Unlike AHL, farnesol which is another signaling molecule, is not sensitive to changes in pH, therefore, the presence of biochar does not have a significant effect on inter- and intra-species signalling by fungi using farnesol [206, 213].

Biochar also possess the ability to adsorb AHL and this adsorption is influenced by the pyrolysis temperature during biochar production. Specifically, biochar produced at higher temperatures through cracking processes has a greater capacity to adsorb AHL molecules. The interaction between biochar and AHL can have implications for gene expression and inter/intraspecific communication among soil microorganisms [214]. This suggests that the presence of biochar in the soil can potentially interfere with the signaling mechanisms of Gram-negative bacteria, impacting their ability to communicate and regulate behaviour. Recently, Li et al. [215] found that biochar amendment significantly enhanced (2.7-fold) the secretion of AHLs, which subsequently promoted the synthesis of C-type cytochromes and humic acid-like substances, improving the electron exchange capability of extracellular polymeric substances (EPS), and thereby promoted direct interspecies electron transfer (DIET). Additionally, increase in the proportion of polysaccharides in EPS due to biochar-induced AHLs enriched potential DIET partners, with the relative abundance of *Geobacter* and *Methanosaeta* increasing from 2.88% and 54.69% to 4.91% and 55.60%, respectively. Thus, enrichment of electroactive microorganisms by biochar-induced AHLs and enhanced DIET improved EPS electron transfer capacity and thereby, enhanced methane production. It was reported that addition of biochar accelerated the degradation of volatile fatty acids, and increased the maximum methanation rate by $24.97 \pm 0.23\%$ in comparison to the control system.

4 Role of biochar in mitigation of abiotic and biotic stresses

Agriculture is the most vulnerable sector, which is often exposed to plethora of climate changes. Global warming, changes in precipitation patterns, recent abrupt changes in climatic conditions and other catastrophes have increased incidences of abiotic and

biotic stresses that manage the operations of the agroecosystem, consequently influencing the worldwide food security [54, 216, 217]. Harsh environments such as drought, salinity, water logging, extreme temperatures (heat, cold and frost), intense light, changes in pH, flooding and heavy metals have been accounted for as major cause for stagnation of productivity in agriculture and horticulture crops [218–221] and cause up to 51–82% losses annually in agricultural productivity worldwide. These abiotic stresses adversely affect the plants at different molecular, physiological, biochemical as well as cellular levels, and impair crop growth, development, survival, yields and food production [6, 222, 223]. On the other hand, biotic stress is caused by biological factors such as diseases triggered by microorganisms (bacteria, molds and viruses), insects, nematodes and weeds [224, 225], which cause huge economic losses in agriculture and forestry by reducing yield and quality. According to a global survey of the major food crops, pests, pathogens and weeds cause average yield losses ranging from 17.2% in potato to 30.0% in rice [226]. Climate changes aggravate the effect of these stresses [54, 227].

Use of naturally-occurring beneficial microbes has emerged as novel cost-effective and eco-friendly strategy to manage environmental stresses in plants to boost sustainable agricultural development [218, 228–231]. Recently, biochar is also gaining attention as a potentially valuable input for agriculture for mitigating the impact of abiotic and biotic stresses and to improve fertility of soil [44, 232]. Biochar have several benefits in improvement of soil health, increasing crop yield, decreasing GHG emissions and mitigation of climate change. It also promotes beneficial microbial community and alleviates impact of drought, salinity and heat stress on the crops [6, 233, 234].

4.1 Biochar as a potential amendment to mitigate drought and salinity stress

One major abiotic stressor that causes a significant decrease in agricultural output is drought stress [6, 235]. Many functions are disrupted by drought stress, ranging from morphological to physiological and anatomical level. Drought stress reduces the production of chlorophyll and causes senescence, which significantly reduces photosynthesis and agricultural yield [236, 237]. According to Vijayaraghavareddy et al. [238], drought stress leads to generation of reactive oxygen species (ROS), which damage DNA, proteins, lipids, and enzymatic processes [239, 240]. Crop productivity declines as a result of detrimental effects caused by drought stress on agronomic attributes and physiological processes [241]. Developing strategies to mitigate the effects of drought stress by biochar amendment is recently gaining attention for ensuring sustainable and resilient agricultural systems that can positively impact soil and plant properties, ultimately helping in mitigating the detrimental effects of drought stress on crops [6, 242]. Thus, biochar is usually referred to as the "black gold" of agriculture, indicating its high value and significance in the context of improving agricultural production.

Application of biochar to soil is associated with several positive effects, including an increase in SOC, soil moisture contents, water and nutrient uptake, cation exchange capacity (CEC), and aggregate stability. These improvements contribute in enhancing drought tolerance in plants [243]. In addition to having a negative impact on plasma membrane, drought stress dehydrates the cytoplasm, which leads to increased electrolyte leakage (EL) and lipid peroxidation [244]. In a study conducted by Hafez et al. [245], the addition of biochar and chitosan contributed to enhanced drought stress tolerance in barley plants. This enhancement is attributed to a decrease in electrolyte leakage and

lipid peroxidation. The improvements in membrane stability, relative water content (RWC), and water pressure are suggested as mechanisms for this increased tolerance. It has been observed that in *Medicago ciliaris* plants grown under drought stress and treated with biochar, there is a significant reduction in the concentration of malondialdehyde (MDA), a marker indicating lipid peroxidation under drought conditions [221]. Yildirim et al. [246] found that biochar decreases MDA accumulation in *Brassica oleracea* by increasing the activity of antioxidant enzymes. This implies that biochar may contribute to reducing oxidative damage by enhancing the plant's antioxidant defense mechanisms.

Osmolytes also play an indispensable role in combating drought stress. It has been observed that drought stress can disturb the hormonal balance and osmolyte accumulation in plants. Proline, produced under stress conditions, is having a dual role as a reactive oxygen species (ROS) scavenger and for osmotic adjustment. This reduction is attributed to less ROS production and reduced oxidative and osmotic stresses in biochar-amended plants. In the case of barley plants treated with biochar and chitosan, there is a marked reduction in soluble sugars, sucrose, and starch contents under stressed and controlled conditions. This suggests an influence on carbohydrate metabolism in response to these amendments [245]. According to Gullap et al. [247], biochar application and irrigation levels significantly affected the contents of abscisic acid (ABA), indole-3-acetic acid (IAA), and gibberellic acid (GA) in soybean plants under drought stress. Biochar application was observed to increase IAA and GA, while decreasing ABA contents and maintain hormone balance. Thus, biochar helps in maintaining hormones and osmolyte accumulation in plants [221]. In a stride to evaluate the individual and combined impact of biochar and plant growth-promoting bacteria (PGPB), olive trees were treated with *Bacillus siamensis* and biochar from olive tree pruning for enhancing resilience to water stress [248]. Increased aerial biomass was obtained in olive trees compared to those treated separately or untreated under both water-stressed and optimal conditions treated with the biochar-bacteria combination. Various stress indicators, including ABA and oxidative damage markers (H_2O_2 and MDA), along with downregulation of dehydrin (DHN) genes (which counteract lipid peroxidation) were reduced. Additionally, downregulation of aquaporin (AQP) genes suggested improved water status by reducing water transport across cell walls, thus aiding cell hydration.

Soil salinity is another critical constraint to crop productivity and global food security [220, 249, 250]. The application of biochar is associated with favorable changes in soil and plant functioning to enhance salinity stress tolerance [251–253]. Various mechanisms are involved by which biochar improves salinity tolerance [221] (Fig. 4). Salinity stress causes membrane damage and this is associated with an increase in MDA accumulation [254]. The concentration and composition of fatty acids and lipids has a significant influence on membrane functioning, stability and fluidity. Unsaturated fatty acids are specifically highlighted for their critical role in membrane protein activity and membrane protection [255]. The application of biochar is reported to increase the concentration of unsaturated fatty acids, which is suggested to improve membrane stability under salinity stress [256]. The application of biochar also improves antioxidant activities, including those of enzymes such as ascorbate peroxidase (APX), catalase (CAT), peroxidase (POD), superoxide dismutase (SOD), and glutathione reductase (GR). These antioxidant activities are reported to protect the membranes from the damaging effects

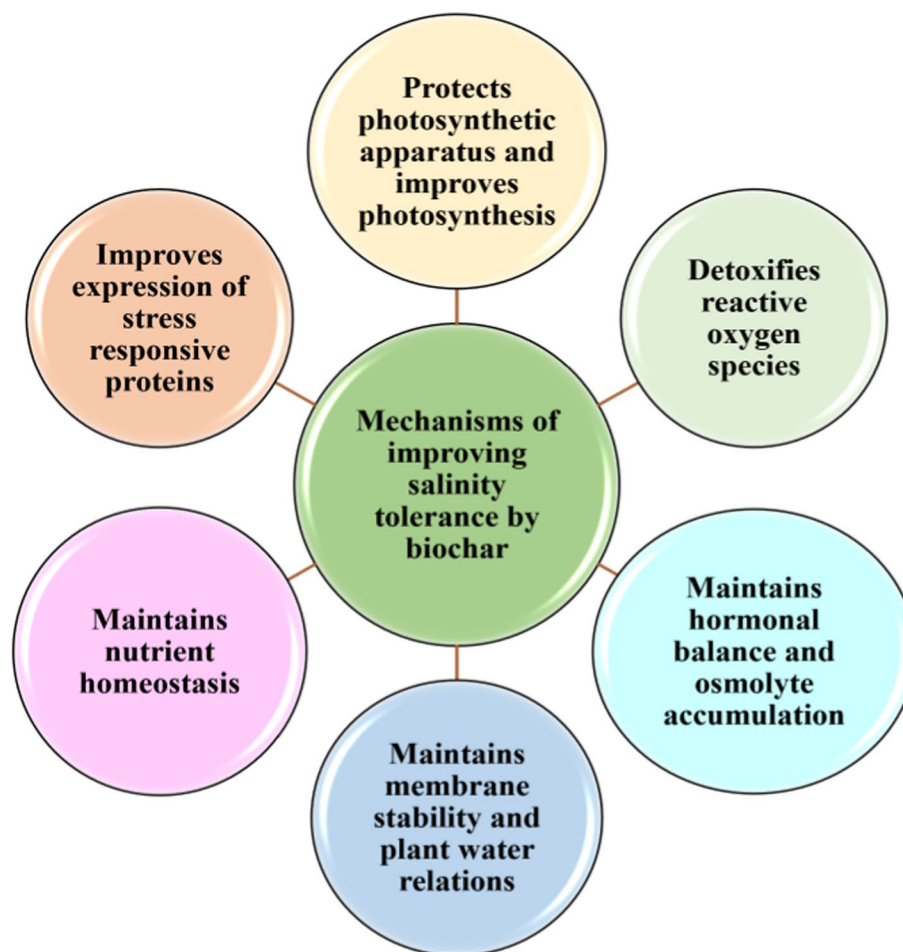


Fig. 4 Mechanisms involved in enhancing salinity tolerance using biochar. Application of biochar to soil is associated with several positive effects that improves salinity tolerance such as improved photosynthesis in plants, detoxification of reactive oxygen species, increase in soil moisture contents, water and nutrient accumulation, membrane stability, and improved expression of stress responsive genes

of salinity stress [257]. Lashari et al. [258] reported that the application of manure and biochar has a positive impact on membrane stability by reducing MDA concentration. A reduction in MDA concentration implies lower levels of lipid peroxidation, indicating a positive impact on the integrity and stability of cellular membranes.

Biochar application also contributes to an improvement in the leaf water status of plants by increasing potassium concentration in plant tissues and improving leaf relative water content (RWC). Osmoprotectants are substances that help protect cells from the adverse effects of osmotic stress, such as water loss or excess salinity [259]. Relative water content is a measure of the water content in plant tissues relative to their maximum water-holding capacity. An increase in RWC indicates improved water retention and hydration in plant leaves. The improvement in leaf RWC after biochar application leads to increase in water use efficiency (WUE) by plants [260]. The application of biochar is reported to promote the uptake of nutrients by plants. This improvement is suggested to contribute to salinity stress tolerance by modifying cellular signaling pathways [261]. In addition, the improvement in Ca^{2+} content is specifically linked to enhanced salinity stress tolerance. According to Kim et al. [257], biochar application is reported to increase tissue P contents. This increase is found to be dose-dependent, suggesting that

higher doses of biochar result in higher P concentrations in plant tissues. Usman et al. [262] found that biochar substantially increased the concentrations of several elements such as P, K, iron (Fe), manganese (Mn), Zn, and copper (Cu) in tomato plants growing under salinity stress. The dose-dependent response highlighted by Kim et al. [257] underscores the importance of considering the application rate of biochar for optimizing nutrient benefits.

In another experiment, Jin et al. [253] analyzed the long-term effects of biochar application in saline-alkali paddy soils and its effects were evaluated on soil physicochemical properties, nutrient availability, soil enzyme activities and rice yield. The application of BC continuously reduced the soil Na^+ concentration, Na^+/K^+ ratio, $\text{Na}^+/\text{Ca}^{2+}$ ratio, exchangeable sodium percentage (ESP), and sodium adsorption ratio. Over the 6-year study, BC amendment also enhanced the concentrations of K^+ , Ca^{2+} , Mg^{2+} , and CEC, which indicated the long-term positive impact of biochar application saline-alkali paddy soils. Interestingly, application of BC at higher doses i.e., 3.0% (T3.0) and 4.5% (T4.5) significantly improved nutrient availability and soil enzyme activities. In similar studies, the combined treatment of BC (prepared from moss) and halotolerant *Halomonas salifodinae* remarkably assisted the root binding and biofilm formation of *H. salifodinae* on the *Medicago sativa* grass roots [66–68]. Moreover, the activity of soil enzymes, including peroxidase, alkaline phosphatase and urease also increased after the combined treatment. Meanwhile, the levels of various cations e.g., Na^+ , Cu^{2+} , Fe^{2+} , Mg^{2+} , Mn^{2+} , etc. in the rhizosphere soil were reduced after the combined treatment, suggesting an improvement in the saline–alkali soil quality.

Recently, Zubairu et al. [263] reviewed the pivotal role of biochar in sustainable remediation of soil salinity. Results showed that biochar amendment remediates soil salinity due to its inherent characteristics like high CEC and possession of anion exchange capacity (AEC) that deprives both cations and anions that form salt in the soil. Application of biochar positively favours the growth of crops under salt stress by boosting K^+ availability, high salt sorption capacity, increased plant water intake, accelerated transpiration, lowered soil EC, increased soil accessible nutrients, increased soil moisture saving, and optimizes the water-salt balance of the soil during saline irrigation, which all favoured crop performance. In another experiment, Jing et al. [264] reported that the simultaneous addition of biochar and *Bacillus fexus* under salt-stress conditions improved the soil physicochemical properties and increased the humus components content and stability, which improved the soil quality of saline–alkali land and enhanced the productivity of fruit mulberry. The salt stress + *B. fexus* + biochar (SS + P + B) treatment significantly increased soil total porosity, non-capillary porosity, pH value, electrical conductivity, the water stable macro-aggregates, organic matter, humic acid (HA) and humin (HM) contents, the HA/fulvic acid (FA) and HA/HE (humus-extractable) ratios, and the activities of catalase and urease.

4.2 Role of biochar in mitigating biotic stresses

Biochar has the potential to suppress the plant diseases caused by soil pathogens [265, 266]. Biochar amendment has shown pathogen-suppressive effects against a variety of plant pathogens, including bacteria, fungi, oomycetes, and nematodes, and has been associated with improved plant growth [66–68, 267, 268]. Biochar have the ability to reduce plant pathogenicity by enhancing resistance, increasing nutritional content and

retention in the soil, improving soil structure and removing toxic compounds from contaminated soils through detoxification and adsorption. For instance, biochar application resulted in increased resistance to specific plant pathogens such as *Leveillula taurica* and *Botrytis cinerea* in tomatoes and pepper [269]. *Leveillula taurica* is a fungus that causes powdery mildew in plants and *Botrytis cinerea* is a fungus that causes gray mold in various crops. The degree/level of plant stress resistance is related to the amount of biochar applied. Higher application rates of biochar resulted in a greater increase in plant stress resistance compared to lower application rates. Biochar amendment can also enhance a plant's resistance to herbivores by affecting the levels of a plant hormone jasmonic acid that plays a key role in defense mechanisms against herbivores. By influencing JA levels, biochar may contribute in making plants less attractive or more resistant to herbivorous pests.

The effectiveness of biochar depends on various factors, including the type of biochar used, soil conditions, and specific plant species. Gao et al. [266] reported that biochar from wheat straw improved the health of tomato plants infected with *Ralstonia solanacearum*, and minimized the severity of bacterial wilt. Also it increased the levels of total C, N, K, P, C:N ratio, pH, and electrical conductivity. Biochar produced from maize has been found to improve the crop response to the pathogens. In a study, maize stalk biochar reduced the effect of *Phytophthora* blight (caused by *Phytophthora capsica*) on pepper up to 50% by increasing diversity and abundance of biocontrol fungi such as *Aspergillus* and *Trichoderma* [270]. Arshad et al. [271] studied the effect of rice husk biochar on tomato plant, which increased the biomass of plant as well as trigger defence related genes (such as PR-1b and JERF3) to reduce the infection of nematode *Meloidogyne incognita*. According to Chen et al. [272], biochar application was found to reduce the reproductive potential of the English grain aphid *Sitobion avenae* in wheat by the activation of defense-related genes in wheat.

Bamboo charcoal is recognized as a regenerative biochar due to its cost-effectiveness and environmental friendliness [273]. The by-products of bamboo charcoal have been reported to show activities against specific insect pests, including *Tetrenchus cinnabarinus*, *Empoasca flavescens*, and *Aphis medicagini* [274]. This suggests that the use of bamboo charcoal may have a dual benefit by improving plant growth and simultaneously providing a level of protection against certain insect pests. Application of bamboo charcoal is said to induce the biosynthesis of flavonoids, terpenoids, and phenolic acids in plants, which are known for their antioxidant and antimicrobial properties, and they can play a crucial role in the plant's defense against herbivores and pests. Thus, use of bamboo charcoal promotes healthier plant growth, and the plant becomes more robust and resilient to external stress pressures. For instance, the induced production of flavonoids, terpenoids, and phenolic acids enhanced the tomato plant's tolerance against destructive pest South American Tomato Pinworm *Tuta absoluta*, commonly known as the tomato leaf miner, which affects tomato plants and other solanaceous crops [275]. These compounds can act as natural defenses, deterring or inhibiting the survival and development of the pest. Thus, application of bamboo charcoal appears to trigger a cascade of biochemical responses in plants, leading to the synthesis of compounds that benefit plant growth and enhance tolerance against pests like *T. absoluta*. This aligns with the broader concept of using biochar and natural plant defense mechanisms for sustainable and environmentally friendly pest management in agriculture.

The combined treatment of biochar with different oil cakes or antagonistic microorganisms decreased the population of pathogens and root-knot nematodes. For instance, amendment with BC and biochar combined with synthetic community (SynCom) consisting of PGPB suppressed soil-borne pathogens and alleviated replant disease in *Radix pseudostellariae* by modifying rhizosphere soil protistan and plant-associated microbial communities [276]. Biochar alone and its combined treatment with *Bacillus* SynCom improved the physiological traits and main agronomic components of *Radix pseudostellariae* under field experiments. The relative abundance of pathogenic *Ralstonia* was significantly decreased with the combined treatment, and it stimulated the *Bacillus* abundance in the root and *Paenibacillus* in the leaf, and the combined treatment also increased the abundance of the predatory protists *Cercomonas* and *Paracercomonas* in the soil. In similar studies, effect of BC alone and in combination with different oil cakes (viz. castor, coconut, mustard, linseed, and sesame) was evaluated on tomato (*Solanum lycopersicum* L.) plants cv. Pusa–Ruby [267]. Biochar in combination with various oil cakes developed resistance to the root-knot nematode *Meloidogyne incognita* in tomato plants. Furthermore, the eggmass/root, the number of galls and nematode population were decreased after BC amendment with oil cakes as compared to the untreated control plants. Higher effect in declining the nematode population was reported with amendment of BC and castor oil cake, followed by treatments with BC + mustard cake, BC + sesame cake, BC + linseed cake, and BC + coconut cake. Minimal effect in declining the nematode population was observed when tomato plants were treated with biochar alone.

Silicon (Si)-modified biochar was reported to significantly reduce the abundance of *Ralstonia solanacearum* in soil by 41.3%, and in tomato roots, stems, and leaves by 67.1%, 72.6%, and 65.1%, respectively [66–68]. The wilt disease index was reduced by 73.4%, indicating effective suppression of soil-borne pathogens. Moreover, Si-modified biochar significantly increased the relative abundances of beneficial taxa such as *Bacillus* and *Xanthomonadaceae* by 482.7% and 52.7%, contributing to enhanced microbial diversity and network stability. Thus, Si-modified biochar mitigated bacterial wilt by enriching beneficial microbes, increasing soil enzyme activity, and promoting C sequestration, collectively reinforcing rhizosphere soil health. In another study, Kumari et al. [277] characterized four *Trichoderma* isolates (*T. harzianum*, *T. asperellum*, *T. virens*, and *T. lixii*) with antipathogenic properties and their GC–MS analysis revealed 37 VOCs. The combined application of *Trichoderma* and biochar (T9) significantly enhanced root length (9.23 cm), plant height (26.03 cm), and root mass (43.33 g) in chickpea plants. Moreover, treatments (T9) and (T10) reduced the disease incidence in chickpeas, decreasing Fusarium wilt by 27% and collar rot by 33%, respectively. Scanning electron microscopy analysis of *T. harzianum* revealed mycoparasitic structures, including hyphal coils, penetration holes, and appressoria, indicating effective pathogen interaction. In similar experiment, the distinct regulatory mechanisms of biochar (BC) on *Bacillus subtilis* and *Rhizoctonia solani* were explored to understand the growth inhibition of pathogens [265]. BC, consisting of micro-BC and nano-BC, caused oxidative stress due to production of ROS. *B. subtilis* secreted EPS, which aggregated nano-BC, enhanced cell adhesion, and reduced intracellular ROS (from 2.0 to 1.5-fold), promoting growth later with BC's nutrient support. Conversely, *R. solani* can not block nano-BC entry, activating mitophagy and suppressing genes like ATP1,2 involved in oxidative

phosphorylation and tricarboxylic acid cycle. This results in ATP deficiency, collapses antioxidant system, raises ROS (from 3.9 to 4.5-fold), decreases cell survival, and leads to cell death. These findings highlight BC's selective microbial regulation and its potential for safe agricultural and environmental use.

5 Role of biochar in soil remediation and mitigation of greenhouse gas emissions

As environmental problems continue to grow due to human activities [278, 279], sustainable and creative ways are needed to tackle issues like soil pollution and GHG emissions. One emerging and promising solution for these environmental issues is use of biochar [280, 281]. The carbon-rich porous structure of biochar enhances its ability to retain water, nutrients, particulate matter, capture potentially toxic elements, contribute in remediating soil pollutants and mitigating the release of GHG gases, and also support microbial activity in soils [282, 283]. Biochar's physical and chemical properties make it an ideal candidate for a wide range of environmental applications, including soil remediation and carbon sequestration.

5.1 Soil remediation with biochar

Recently, development of novel materials and innovative technologies has gained more attention to improve the quality of soil to remediate soil contaminants. Heavy metals (arsenic, cobalt, chromium, copper, zinc, mercury, lead and nickel) [284], organic compounds, and other hazardous pollutants are being added in soil due to ineffective waste disposal methods and injudicious use of chemical fertilizers and pesticides (Fig. 5) leading to decrease in soil fertility, reduced crop yield and disturbance in natural microflora distribution. These pollutants may have poisonous, mutagenic, or carcinogenic, characteristics and therefore, may lead to mutations in the genome of soil-inhabiting microbes, plants and animals that absorb them [285, 286]. Current soil remediation techniques encompass physical, chemical, and biological treatments. Excavating waste material, burning them or leaching are examples of physical remediation, however, these methods are costly and may result in secondary problems. Chemical treatments employ reduction, immobilization, and modification; yet, they have the potential to pollute the soil. Compared to physical or chemical treatments, bioremediation approach employing microorganisms and plants has least adverse consequences [287, 288]. Therefore, more economical and less time-consuming remediation measures are needed to lessen adverse side effects of soil contaminants and biochar can be quite helpful in these events [285].

Biochar amendment has proved an effective way to improve soil properties and decrease the harmful effects of soil acidification [127, 128]. It is commonly recognized that adding biochar to soil may bind soil particles and plant roots, increasing both phytoremediation and bioremediation through their interactions [289]. Some studies have also suggested that biochar can adsorb most soil contaminants (organic as well as inorganic) from contaminated soil depending on specific biochar properties determined by the type of biomass used and the pyrolysis temperature applied [283, 290]. Previously, the main focus of adding biochar to soil was not on soil contaminant remediation, but rather on how well it might improve the physical, chemical, and biological properties of the soil [291]. But with recent advances, many studies are focusing on the use of biochar for remediation of soil contaminants [283, 292, 293].



Fig. 5 Different problems occurring in soil due to various anthropogenic activities. Excessive and injudicious use of chemical fertilizers causes environmental pollution problems. However, the reduction in use of chemical fertilizers may often reduce actual agricultural production. Anthropogenic activities cause different problems including heavy metals contamination, nutrient depletion and addition of organic pollutants. On the other hand, amendment of biochar in the soil improves soil structure and serves as a supportive habitat for colonization of beneficial microbial communities that enhance nutrient availability and nutrient use efficiency, resulting in reduced fertilizer dependency of crop plants

Due to extensive agriculture, soil is being contaminated by many HMs such as Cd, Cr, Hg, Pb, Zn, Cu, Ni, Se, As and Co, and persistent organic pollutants (POPs) [such as polychlorinated biphenyls (PCB), polycyclic aromatic hydrocarbons (PAH) and polychlorinated dibenzofurans] [284]. Recent studies suggest that BC application in such contaminated soils immobilizes HMs and POPs by different methods [294]. The in situ application of BC in contaminated soils bind pollutants, promote plant growth and stimulate ecological diversity [289, 295]. It may also act as an important sink for dissolved pollutants in soil [296, 297]. Besides this, BC can also manipulate soil acidity by increasing soil pH by mineralizing their innate ash alkalinity and sequestering the SOC

for chelating the toxic Al^{3+} . The negative charge due to the functional groups on the surface of biochar is responsible for attracting the positively charged pollutants [233]. For instance, applying 5% rice-straw biochar in cadmium-contaminated soil (50 mg kg^{-1}) led to a 19.3% reduction in Cd levels, whereas bamboo biochar achieved an 8.6% reduction [298].

In a recent study, Xiong et al. [55] conducted a pot experiment using kitchen-waste biochar prepared at 300, 500, and 700 °C, which was applied either alone or in combination with *Escherichia coli*. Low-temperature biochar (300 °C) exhibited limited Cd immobilization, reducing the acid-extractable Cd fraction by only $4.22\% \pm 0.20\%$ compared with the control, but markedly enhanced soil fertility. In contrast, high-temperature biochar (700 °C) reduced the acid-extractable Cd fraction by $28.34\% \pm 0.50\%$ and the total bioavailable Cd (F1 + F2) by $22.25\% \pm 0.04\%$ relative to the water-treated control. The enhanced Cd stabilization of the 700 °C biochar was attributed to its well-developed pore structure and high alkalinity, which created favorable habitats for beneficial taxa such as *Bacillus*, *Rhodococcus*, and *Mucor* while suppressing competitors. Although *E. coli* alone had negligible effects on Cd mobility, its co-application with low-temperature biochars even increased Cd bioavailability and plant uptake, whereas combination with the 700 °C biochar strongly immobilized Cd and minimized translocation risk. These findings suggested that the temperature-dependent physicochemical properties of biochar influence microbial communities, contributing to enhanced Cd immobilization in soil and provided a new theoretical basis for the selective application of biochar and efficient Cd immobilization in HM-contaminated soils. Biochar contributes to soil remediation by different mechanisms, which includes adsorption of pollutants, cation exchange capacity, pH buffering and particle size of biochar (Fig. 6).

1. Adsorption of pollutants: One of the primary mechanisms through which biochar contributes to soil remediation is adsorption. Biochar's porous structure provides a large surface area for adsorption, effectively trapping and immobilizing a variety of pollutants such as HMs, pesticides, and organic contaminants. This prevents these pollutants from leaching into groundwater or being taken up by plants thereby reducing the risk of environmental contamination [299, 300].

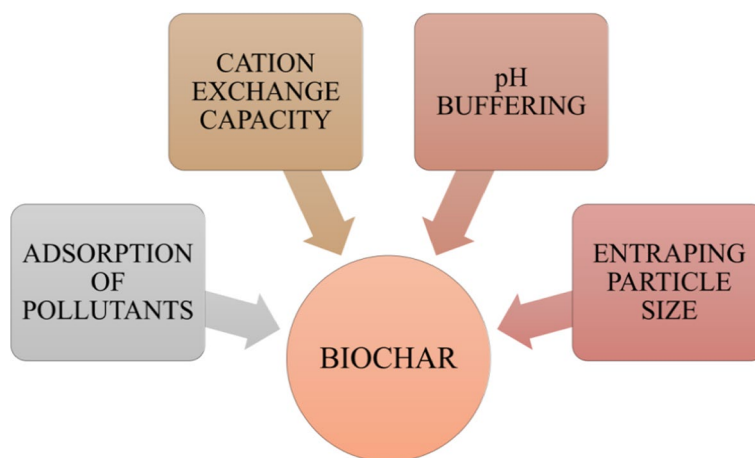


Fig. 6 Biochar amendment contributes to soil remediation by different mechanisms, which includes adsorption of pollutants, cation exchange capacity, pH buffering and particle size of biochar

2. Cation exchange capacity: Biochar enhances the soil's cation exchange capacity, which refers to its ability to retain and exchange essential nutrients. This property is crucial for nutrient-poor soils, as it helps improve fertility and promotes healthier plant growth [301]. Additionally, the increased CEC can contribute to the immobilization of certain contaminants, further aiding in soil remediation [302].
3. pH buffering: Biochar also acts as a pH buffer, helping to maintain optimal pH levels in soils. This is particularly beneficial in acidic or alkaline soils where biochar can neutralize acidity or alkalinity, creating a more favorable environment for plant growth. Proper pH levels are essential for the availability of nutrients and the overall health of the soil ecosystem [303].
4. Particle size: Another important parameter of biochar for soil remediation is its size. The size range of BC utilized as soil amendment is typically between <0.10 and 4 + mm; however, in a meta-analysis study of plant responses conducted by Thomas [304], it was suggested that biochar with particles sized at an intermediate range of ~0.5–1 mm was ideal for soil remediation. This relatively small particle size and high porosity provides biochar with great specific surface area and cation exchange capacity enabling it to retain water, nutrients and pollutants effectively [305]. Muhamad et al. [306] reported that the addition of hardwood biochar significantly increased the mineralization of ^{14}C -glucose at 5% biochar dosage and at finer particle sizes (2, 1, 0.5 and 0.1 mm). The pH of soil and carbon, and the microbial biomass also increased after biochar addition in unfertilized soil. Adding biochar to soil had no major change on the leaching of nitrate ions, but reduced the ammonium ion leaching. The efficacy of biochar application depends on soil type, nutrient availability, biochar application rate and particle size.

The combined application of rice husk BC (RH; at 0%, 2.5%, and 5% doses) and foliar application of salicylic acid (SA; 0, 0.5, and 1 mM concentration) on Chinese mustard caused greatest reductions of HMs such as Pb, Cu, Cd, and Zn concentrations i.e., 74.72, 69.19, 54.16, and 53.86%, respectively in the shoots as compared with untreated control plants [307]. Lu et al. [308] amended the six acidic paddy soils with seeded sunflower plate biochar (SSPBC), peanut straw biochar (PSBC) and corn straw biochar (CSBC). The incorporation of crop residue biochars led to the increases in soil pH and pHBC, but decreased soil available Cd content. The decreasing order of available Cd content was SSPBC > PSBC > CSBC and was consistent with the changes in soil pH induced by different kind of biochars. Also, there was a significant negative correlation between the change in available Cd content and soil pHBC during submerging/drainage alternation. Consequently, the amount of Cd absorbed by rice in Cd-contaminated paddy soils was reduced, thereby reducing the potential risk of soil Cd to humans. In another study, Gong et al. [309] used mixed BC derived from rice husk and cow bone (MB), and it was loaded with cadmium (Cd)-tolerant PGPB to assess its performance on radish in a pot experiment. Results showed that MBM effectively reduced soil diethylenetriaminepentaacetic acid extractable Cd content by 58.61% and Cd content in radish roots were decreased by 73.48%. MBM also significantly increased the abundance of phosphorous cycling functional genes, plant growth and soil health.

In a recent experiment, Chang et al. [310] developed a biochar-loaded *Ac* material and clarified its chemical and microbial mechanisms for Cd immobilization and growth promotion of lettuce plants. It was reported that biochar-loaded nitrogen-fixing

bacteria (*Azotobacter chroococcum*; BAc) enhanced Cd adsorption by forming stable complexes with bacterial secretions and activating biochar functional groups. Compared with biochar and Ac, Ac successfully colonized the lettuce rhizosphere, tagged with green fluorescent protein after BAc application. BAc reduced soil Cd bioavailability by 16.67–46.42% and Cd accumulation in root and shoot by 14.28–69.74%. This occurred by increasing soil pH and converting exchangeable Cd to carbonate-bound and Fe/Mn oxide-bound fractions. In another study, the synergistic effects of rice biochar (RBC) and *Metabacillus indicus* AMB4 (PGPR) on arsenic (As) bioavailability and rice physiological responses were evaluated under controlled and field conditions [73]. Combined RBC + PGPR treatment reduced soil As bioavailability by 85% and plant As accumulation by 78% as compared to control treatments, while significantly enhanced antioxidant enzyme activities (CAT: 4.2-fold, SOD: 3.7-fold, APX: 3.1-fold increase). Metagenomic studies revealed restructured microbial communities with increased Proteobacteria abundance and upregulated As detoxification genes (*arsC*, *arsB*, *aioA*) coupled with enhanced nutrient cycling pathways (*nifH*, *phoD*, *soxA*). Network analysis demonstrated functional integration between As-transformation and nutrient metabolism, with coordinated expression of 47 arsenic-responsive and 126 nutrient-cycling genes. In another study, Chauhan et al. [311] reviewed the pivotal role of microbial communities in facilitating the adsorption, transformation, detoxification of toxic heavy metal (THMs), through enzymatic activities and production of secondary metabolites, such as metalloproteins, siderophores and exopolysaccharides, and others. Additionally, synergistic application of biochar with beneficial microbes enhanced the THMs removal efficiency, minimized toxicity and promoted plant health, thus contributing to environmental protection and sustainable agriculture.

Biochar amendments have also been found to influence pesticide degradation. Temperature significantly impacts adsorption efficiency of pesticides. Increasing pyrolysis temperature from 250 °C to 850 °C raised atrazine sorption on maize-stalk biochar from 21.9 to 35.8 mg g⁻¹ due to a larger surface area and more functional groups [312]. The half-life of flubendiamide decreased from 165 days to 103 and 117 days with 5% and 10% biochar amendments, respectively, due to biochar's adsorption limiting microbial degradation [313]. Additionally, biochar modified with nanoparticles exhibited a significantly higher sorption of pesticides capacity of 147 mg g⁻¹, compared to unmodified biochar, which had a capacity of 67.8 mg g⁻¹ [314]. This enhancement can be attributed to the nanoparticles, which increased both the surface area and the functional groups present on the biochar. This confirms that biomixtures perform well in comparison to biochar alone with reference to pesticides sorption [315].

The influence of BC amendment was also investigated on microbial community succession in soil contaminated by PAH-phenanthrene and on mitigation of the toxicity effects on soil microbes [316]. The biochar-amended treatments (BC) and biochar immobilized with the phenanthrene-degrading bacteria (MBC) showed significantly lower phenanthrene removal rate (PRR) than the control treatments lacking biochar application or without inoculation (CK). Bacterial inoculation showed negligible effect on the soil bacterial community, whereas the BC amendment significantly changed the bacterial community structure and strengthened the bacterial co-occurrence network. Additionally, soil dehydrogenase activity (SDHA) and leaf CYP450 (cytochrome P450) activities were significantly higher in the BC, MB, and MBC treatments and lower than those in

the CK treatment, respectively. Furthermore, BC amendment alleviated the toxicity or stress of PAH on soil microbes. In another study, the mechanisms of biochar-assisted di-2-ethylhexyl phthalate (DEHP) degradation were explored in organic contaminated farmland soils. The amendment of biochar (at 1.0% concentration) caused a significant increase in rhizosphere pH, OM and humus, and achieved a satisfactorily higher DEHP removal (maximally 77.53%) efficiency [317]. Biochar addition also remarkably changed rhizosphere bacterial communities by enriching some potential DEHP degraders including *Nocardioides*, *Sphingomonas*, *Bradyrhizobium* and *Rhodanobacter*. The abundance of genes encoding key enzymes (hydrolase, esterase and cytochrome P450) and DEHP-degrading genes (*pht3*, *pht4*, *pht5*, *benC-xylZ* and *benD-xylL*) were increased after BC amendment, leading to the change in DEHP degradation metabolism, primarily from benzoic acid pathway to protocatechuic acid pathway.

He et al. [318] immobilized microorganisms on biochar to regulate the electron transfer capacity for anaerobic digestion (AD) of high-load organic wastewater. Compared with uncoupled groups, the CH₄ yield of pre-coupled groups increased by 26.4–36.9%, the lag phase shortened from 4.45 to 5.62 days to 0.85–1.06 days, and soluble chemical oxygen demand removal efficiency increased from 60.8 to 63.8% to 92.4–96.4%. Pre-coupling enhanced microbial activity and abundance, reduced the bacteria-archaea spatial distance, and promoted direct interspecies electron transfer. Synergistic microbes (*Syntrophobacter* and *Syntrophomonas*) were enriched, potentially establishing syntrophic relationships with methanogens to promote organic degradation. The developed biochar-microorganism hybrid demonstrated the potential to reduce hydraulic retention time and improve AD efficiency, offering valuable guidance for practical applications. Recently, Ghaedi et al. [319] reviewed the potential applications of metal organic frameworks (MOF) based composites in environmental engineering and pollution control. The adsorption capacity of composites is generally doubled compared to the standalone biochar, while MOFs maintain their crystallinity, even over multiple regeneration cycles, indicating the composites' long-term applicability and sustainability. These findings highlight the potential of MOF-biochar composites for environmental applications and identify key areas for further research to enhance their sustainability in environmental protection and green energy.

Peng et al. [320] developed an innovative solution for waste activated sludge (WAS) management in which WAS was converted to biochar and feedback catalyzed biosynthesis of polyhydroxyalkanoates (PHA). Biochar significantly improved electron transfer during anaerobic digestion (by 61.36%) and boosted the proportion of PHA-producing bacteria by 12.33%, resulting in an 81.04% increase in PHA yield, comprising 66.14% of the dry sludge weight. Multi-omics analysis showed that biochar amplified the expression of key PHA synthesis genes (*phaA*, 4.69-fold; *phaB*, 10.04-fold; and *phaC*, 12.02-fold). Thus, a sustainable approach was developed for WAS management, thereby advancing resource efficiency and providing ecological benefits. In a recent study, Zhang et al. [321] synthesized a novel rape straw-derived biochar-calcium alginate composite (M-CA-RBC) with immobilized *Pseudomonas* sp. H6 to simultaneously remove phosphate (PO₄³⁻) and ammonium (NH₄⁺) from distillery wastewater. The results showed that M-CA-RBC had good removal performance for PO₄³⁻ (17.81 mg g⁻¹) and NH₄⁺ (25.78 mg g⁻¹). After adsorption saturation, M-CA-RBC exhibited good stability, which could be used as a slow-release fertilizer to promote the growth of mung bean plants.

5.2 Mitigation of greenhouse gas emissions by application of biochar

Agricultural production systems currently account for 10–14% of GHG emissions, but by 2030, it is predicted to increase upto 50% [322, 323]. Therefore, implementing a vigorous climate change mitigation program is essential for achieving environmental sustainability. For this perspective, future agricultural production systems must be carbon neutral or negative in order to lessen the consequences of GHG emissions and global warming [75, 324, 325]. The "4 per 1000" campaign, which focuses on soils for food security and the environment to solve global GHG emission concerns, was started by the United Nations Framework Convention on Climate Change [326]; <https://www.4p1000.org>. It aims to increase the yearly SOC sequestration of agricultural lands globally to 2–3 Gt C in the upper 100 cm soil in order to successfully offset 20–35% of the global C production. Therefore, high food-producing carbon negative technologies must be applied to address the numerous challenges [75]. Use of biochar has a potential as a cutting-edge technique to alleviate environmental issues and food hunger by increasing agricultural production, lowering GHG emissions, and minimizing soil and water pollution [75, 327]. Biochar application in the soil results in substantial reduction of greenhouse gas emissions and various factors such as substrate used and preparation method used in biochar production, its interaction with other inputs and microbial communities, and crop management practices have been found to affect mitigation potential of GHG emissions using biochar in the soil (Fig. 7).

Methane (CH_4) is an important greenhouse gas and has a global warming potential roughly 20 times higher than CO_2 , and studies have demonstrated that biochar can significantly mitigate CH_4 emissions from sources such as rice paddies, livestock manure, natural wetlands and landfills. Biochar supplementation to soil inhibits CH_4 production and lowering of CH_4 level in the atmosphere can contribute to mitigating of climate change. Biochar application initially does not directly inhibit methanogenic bacteria rather changes in the aeration properties of the soil may lead to a decrease in the activity and abundance of methanogenic bacteria. Conversely, there is an increase in the activity and abundance of methanophilic bacteria. Methanophilic bacteria are methane-consuming bacteria that utilize CH_4 as an energy source. These changes directly affect CH_4 emissions [328, 329].

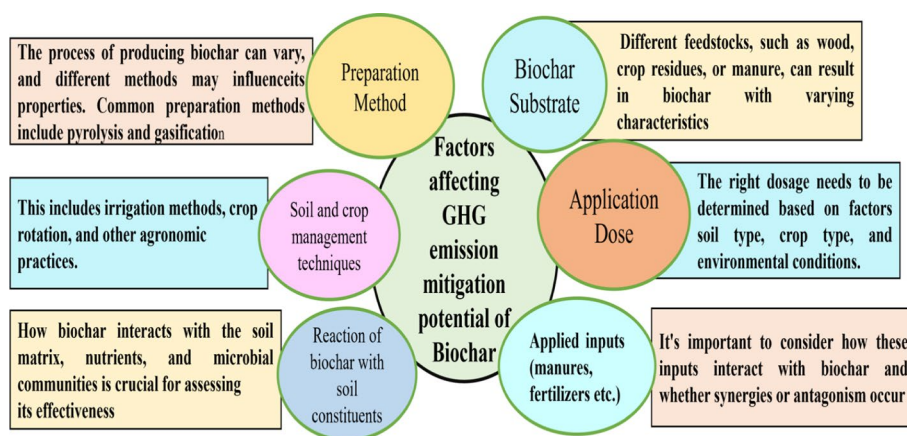


Fig. 7 Factors affecting greenhouse gas emission mitigation potential of biochar. Various factors such as substrate used and preparation method used in biochar production, its interaction with other inputs and microbial communities, and crop management practices affect mitigation potential of GHG emissions in the soil

Another mechanism behind this reduction of CH₄ emission level is its adsorption onto biochar surfaces, followed by microbial oxidation [233]. A long-term field study reported that applying spruce-wood biochar at a rate of 10 t ha⁻¹ led to a 43% decline in CH₄ emissions [330]. In rice paddy fields, biochar derived from rice straw has demonstrated its capacity to lower CH₄ emissions even under elevated CO₂ and temperature conditions [331]. This is achieved by inhibiting methanogen activity while fostering methanotrophic microbial populations [233]. Pyrolysis conditions also play a role in CH₄ emissions i.e., with biochar produced at 300 °C increased CH₄ emissions by 38%, while that produced at 700 °C resulted in an 18.2% reduction in CH₄ emissions [332]. These findings highlight the importance of pyrolysis conditions in determining biochar's impact on GHG emissions [333]. Applying biochar at a rate of 10% (w/w) can greatly reduce the CH₄ emissions. By enhancing soil aeration, decreasing denitrification, and boosting the capacity of the CH₄ sink, biochar helps in retardation of CH₄ release [334]. In soybean crops treated with biochar grown in the acidic soil of the Eastern Colombian Plains, there was an almost complete inhibition of CH₄ emission [335]. Findings in another study suggested that applying biochar in smaller annual doses over four years lowered CH₄ emissions by 41%, whereas a single high-dose application achieved a 38.3% reduction [336]. These results indicated that regular moderate applications of biochar enhance methanotroph activity more effectively than a single large dose, making them a more practical and sustainable strategy for improving soil health.

According to Dawar et al. [337], biochar also affects the proportions of N-cycling in the soil by increasing the rates of ammonia (NH₃) adsorption, nitrification, and ammonia storage by increase in CEC of soil. This alters the efficacy of N input into the soil system. Nitrous oxide (N₂O) is the third abundant greenhouse gas, and it can persist in the atmosphere for 110–180 years. Although N₂O emission in 2020 accounted for about 7% of total greenhouse gases (GHG), its global warming potential is 265–310 times higher than CO₂, resulting in a much stronger greenhouse effect than anticipated. Studies suggest that biochar can reduce N₂O emissions by up to 50% in sandy-loam soils across varying moisture conditions [338]. Liao et al. [339] reported that treatment of oilseed rape (*Brassica napus* L.) with biochar-based controlled release nitrogen fertilizers (BCRNFs) stimulated nitrification by increasing the copy number of the bacterial *amoA* gene and reduced N₂O emission by limiting the abundance of *nirS* and *nirK* as compared to urea treatment. A meta-analysis further reported an average 26% reduction in N₂O emission in biochar amended soils, with the most pronounced effects observed in nitrogen-rich soils [340]. Manure application alone raised N₂O emissions by 17.7%, but combined manure and biochar treatment reduced the emissions by 19.7% [340]. Factors such as the carbon to nitrogen (C:N) ratio, feedstock composition, and pyrolysis temperature, time of application and exposure limit all contributed to biochar's effectiveness in mitigating N₂O emissions [341]. Another study demonstrated that biochar enhances denitrification, reducing N₂O accumulation by up to 98%, primarily due to its bulk structure rather than soluble compounds [170–172]. These findings suggested that biochar creates favourable physical and chemical conditions for denitrifying bacteria. However, its effectiveness in mitigating N₂O emissions depends on the feedstock used and the pyrolysis temperature.

In a recent comprehensive 3-year field study performed in a subtropical forest soil, Zhou et al. [342] revealed that application of straw biochar in the soil resulted in a

substantial reduction of N_2O emissions, ranging from 8.0 to 20.8% each year. Notably, the mitigation effect of biochar was particularly pronounced when N deposition was occurring, leading to an even greater reduction in N_2O emissions by 14.2–22.0% annually. This mitigation effect is attributed to biochar's capacity to lower the nitrification and denitrification rates of soil via reducing levels of ammonium N and water-soluble organic N. Additionally, biochar decreased the abundance of critical microbial genes, including AOA *amoA*, *nirK* and *nirS*, and reduced the activity of key enzymes such as nitrate and nitrite reductase. These findings highlighted the potential of straw biochar to effectively mitigate soil N_2O emissions in subtropical forests experiencing N deposition, offering important insights for supporting ecosystem sustainability under global climate change.

In another study, physical properties of biochar particularly particle size was demonstrated to significantly impact GHG emissions. For instance, smaller biochar particles (0.5–2 mm) are more effective in reducing carbon monoxide (CO) and N_2O emissions than larger ones (5–10 mm) [343]. This is largely due to their increased surface area, which facilitates microbial interactions and accelerates decomposition. In other study, MnO_2 -modified biochar (MBC) was reported to improve microbial denitrification at low temperatures in cold climates [344]. The modification of biochar with MnO_2 considerably raised the specific surface area, pore volume, and electron transfer capacity of BC by 185%, 88% and 30%, respectively. The activities of key enzymes linked to glycolysis (HK, PFK, PK) and denitrification (Nar, Nir, Nos), alongside the electron transport system activity, as well as ATP and NADH levels, were significantly enhanced. The enrichment of *Acidovorax* and *Tolomonas* indicated that microbial manganese redox cycling played a critical role in promoting nitrogen removal. Research also indicated that the reduction of NH_3 , N_2O , and CH_4 emissions is significantly impacted by biochar produced from woody feedstocks and agricultural waste [345]. The impacts of different GHGs on climate and the role of biochar in reducing the GHGs emissions are discussed in Table 3.

Thunuguntla et al. [355] reported that poultry litter and switchgrass biochars produced at 350 and 700 °C (PLB350, PLB700, SGB350 and SGB700, respectively) affected CO_2 conversion to C2–C6 alcohols and acids by *Clostridium muellerianum* P21, *C. ragsdalei* P11 and *C. carboxidivorans* P7. SGB350 increased alcohol titers by 1.1–2.1 fold, and PLB350 enhanced acid concentrations by 1.2–1.7 fold compared to the control without biochar. About 2.0–3.3 fold more ethanol was formed by strain P11 compared to strains P7 and P21 with SGB350. However, strain P21 produced 2.4-fold more butanol than strain P7 with SGB350, including unique hexanol production. These results highlighted the potential of biochar in enhancing C2–C6 alcohol production from CO_2 , thereby boosting process feasibility. Olivera-Begu  et al. [356] studied the effect of the addition (10% DW ratio) of wood-derived biochar on the composting of source-selected organic fraction of municipal solid waste (OFMSW). Addition of biochar significantly decreased methane (from 0.17 to 0.05 kg CH_4 Mg^{-1} OFMSW) and ammonia (from 0.57 to 0.35 kg NH_3 Mg^{-1} OFMSW) emissions over the 80 days of composting. The differences in VOC emission from the two treatments were only significant during the maturation phase (from day 50 onwards) with average values of 35 and 175 g C-VOC day^{-1} , for the biochar and control treatments, respectively. Odor emissions were smaller in the

Table 3 Role of biochar in mitigating greenhouse gases in environment

Green house gas	Impact on climate	Mitigation by biochar	References
Carbon dioxide	Among all the GHGs, CO ₂ has the least global warming potential. However, CO ₂ is the most significant GHG contributing to global warming due to its greater atmospheric concentration and lengthy atmospheric life cycle. It accounts for around 72% of the overall warming brought on by human activity [346]	Biochar plays an important role in C sequestration by locking away carbon in a stable form. When applied to soil, biochar persists for an extended period (because it is resistant to biotic and abiotic decay), preventing the release of CO ₂ into the atmosphere. Biochar's estimated C-residence period in soils can range from hundreds to thousands of years [347]. This not only helps combat climate change by reducing net C emissions but also enhances soil C stocks, contributing to long-term soil health.	[63, 64, 348–350]
Methane	It has a 28-fold greater potential to cause global warming than CO ₂ . It accounts for almost 15% of all climate change and global warming caused by human activity. Biological sources account for around 70–80% of methane emissions, with the remainder coming from natural sources such as forests, swamps, oceans, marshes etc.	Biochar amendments have been shown to reduce CH ₄ emissions by altering microbial pathways and promoting conditions that inhibit methanogenesis. Increased methanotrophic activity, particularly the methanotrophic <i>pmoA</i> gene, promotes rhizospheric activity for methanotrophs, and was identified as the cause of the decrease in CH ₄ emissions in biochar-amended soil [351]. This is particularly relevant in paddy fields [352] and other waterlogged environments.	[63, 64, 348–350]
Nitrous oxide	With a 265-fold greater global warming potential (GWP) than CO ₂ , N ₂ O is a powerful greenhouse gas [346]. It accounts for around 5% of all global warming caused by humans. The main man-made sources of N ₂ O emissions include agriculture, the burning of fossil fuels and industrial processes, biomass burning, atmospheric deposition, and human sewage. Agriculture is the greatest anthropogenic source, accounting for 67% of all anthropogenic N ₂ O emissions. Of these, 25% are indirect emissions from fertilizer runoff and leaching, and 42% are direct emissions from the use of nitrogenous fertilizers and manure management.	Biochar application has demonstrated the potential to reduce N ₂ O emissions because of its high porosity, pH, and specific surface area. Also, by influencing soil microbial processes and nutrient cycling, biochar helps in creating conditions that limit the production and release of N ₂ O. The application of biochar in soil reduced the NO ₃ ³⁻ and NH ⁴⁺ concentration (substrates for nitrification and denitrification reactions) resulting in decreased N ₂ O emissions [353]. In addition, biochar application in soil may improve N ₂ O reductase concentration leading to conversion of N ₂ O into N ₂ , hence, reduced N ₂ O emission [354].	[63, 64, 348–350]

presence of biochar ($1.1\text{E} + 0.6$ vs. $1.9\text{E} + 0.6$ ou Mg^{-1} OFMSW). The agronomic value of the resultant biochar co-compost was similar to that of the control compost.

The performance of biochar, however, depends greatly on the choice of feedstock and the conditions under which it is produced. Biochar made from wood at high temperature are particularly effective for long term C storage, though they offer limited benefits for crop productivity whereas low-temperature biochar derived from nutrient-dense materials such as animal waste can significantly increase yields, especially in nutrient deficit soils. By improving water retention, enhancing nutrient availability, and long-term soil resilience, biochar has emerged as a promising approach towards climate change adaptation and mitigation [357]. Similarly, there are certain factors that affect the role of biochar in mitigating GHG emission (Fig. 7). By manipulating these factors, the biochar efficiency in reducing GHGs and climate change mitigation can be enhanced.

6 Effect of biochar amendments in improving plant growth and crop yield

For minimizing environmental problems caused by excessive application of chemical fertilizers, reduced use of chemical fertilizers is advocated, which may often reduce actual agricultural production. Biochar has been demonstrated to regulate soil nutrient distribution due to its unique properties and significantly affect soil microbial community structure/functions. Therefore, biochar amendment in soil has emerged as a promising technology with the ability to improve soil properties, and increasing plant growth and crop yield in contaminated and problematic soils by boosting populations of beneficial microorganisms [65, 72, 358, 359]. In this section, the contribution of biochar in improving plant growth and crop yield in problematic soils is discussed. To accomplish this, biochar enhances the availability of certain macro- and micro-nutrients to plants by different mechanisms (Tables 4 and 5). In a recent study, Palansooriya et al. [77] reviewed the role of biochar-based controlled-release fertilizers (BCRFs) in enhancing nutrient use efficiency and mitigating the environmental impact of fertilizer use in cropping systems. The BCRFs also improved soil structure and enhanced microbial activities and root growth in productive crops. BCRFs also have potential for C sequestration, mitigation of GHG emissions, reduction in nutrient leaching and environmental impact, contributing to sustainable agricultural practices compared to the use of chemical fertilizers. Thus, BCRFs offer a promising pathway for improving nutrient management in agriculture, however, interdisciplinary efforts are needed to unlock their full potential in enhancing plant growth and environmental sustainability.

Biochar can impact nutrient availability by altering soil pH and cation exchange capacity. It can adsorb and retain nutrients, preventing leaching into surrounding environment and making them more available to plants over time. Biochar can enhance plant nutrient uptake efficiency, photosynthesis efficiency, chlorophyll content, and overall plant growth and yield. The biochar's functional groups, such as hydroxyl, carboxylic, ketone, chromene, and lactone groups strongly adsorb nutritional ions and minimize losses [181]. The effects of biochar on the availability and immobilization of nutrients in the soil varies and depends on the chemical composition of biochar, application dose, soil pH, nutrient status of soils, and microbial interaction with biochar [315]. Applications of biochar reduced NO_3^- leaching; this is because biochar has alkyl and hydroxyl functional groups. Mukherjee et al. [371] provided evidence in support of this claim, and reported a 33% decrease in NO_3^- leaching upon applying biochar at a rate of 10 g kg^{-1} of

Table 4 Understanding the mechanisms of biochar’s role in nutrient availability and its impact on plant growth

Mode of biochar action	Description	References
Nutrient retention and slow release	Biochar’s porous structure offers a large surface area for nutrient adsorption, increasing the availability of essential nutrients for plants over time. During the plant’s growth cycle, this slow-release mechanism helps to maintain nutrient availability	[360, 361]
Enhancement of cation exchange capacity	By increasing CEC, biochar helps to retain these cations in the soil, preventing nutrient runoff and improving the overall nutrient-holding capacity of the soil	[189, 315, 362]
pH modulation	In acidic soils, biochar acts as a pH buffer, raising the pH levels and making the soil more conducive for plant growth. This pH adjustment can unlock certain nutrients that may be less available in more acidic conditions, such as P and micronutrients like Zn and copper	[198, 199, 363]
Improved soil structure and aeration	The physical characteristics of biochar are essential for improving soil structure. Because of its porous structure, the soil has a network of open spaces that improve water retention and aeration. This is favorable for strong plant development, which permits improved root penetration and makes it easier for water and nutrients to reach plant roots	[125, 126, 364]
Water retention and drought resistance	Porous structure of biochar increases plant water access. This is especially helpful in areas that are vulnerable to drought or have unpredictable rainfall patterns. Biochar’s enhanced ability to retain water helps plants stay properly hydrated, which fosters resilience in times of water scarcity	[365, 366]

Table 5 Effect of biochar on release of specific nutrients in soil

Nutrients	Effect of biochar	References
Nitrogen	Biochar indirectly affects N availability in soil by influencing nitrogen cycling Biochar enhances the retention of ammonium ions, reducing the potential for nitrogen leaching It can also stimulate the activity of nitrogen-fixing bacteria	[367–369]
Phosphorous	Biochar's ability to adsorb and retain phosphorus can significantly improve its availability to plants	[370]
Potassium	The porous structure of biochar acts as a reservoir for potassium ions, preventing their leaching and making them available for plant uptake	[370]
Micronutrients	Biochar can also influence the availability of micronutrients (such as Zn, Fe and Cu) in soil The increased CEC and altered pH conditions created by biochar can enhance the solubility and uptake of these micronutrients by plants This is crucial for addressing micronutrient deficiencies, which can limit crop productivity in many regions	[369]

silty loam soil. The addition of biochar also decreased the concentration of NH_4^+ in soil [181]. The amount of NH_4^+ available for nitrification can decrease NH_4^+ sorption to the oxygenated carbonyl and carboxyl functional groups of biochar. Increased nitrate concentration in the soil following incremental dosages of biochar has also been reported [372].

Nguyen et al. [373] described that when urea was applied along with high cation exchange capacity biochar, there was a decrease in nitrification and an increase in the abundance of genes encoding ammonia monooxygenase (*amoA* gene), but not as much in low CEC biochar. Martin et al. [374] provided support for this claim, the addition of biochar increased the activity of N immobilizing heterotrophs, hence decreasing nitrification. In contrast to increasing soil nitrification and *amoA* gene abundance, biochar-based controlled release N-fertilizers decreased denitrification and the abundance of nitrate reductase genes (*nirS* and *nirK*) [339]. On the other hand, it was noted that adding biochar accelerated the denitrification process, which reduced soil N [375]. After applying biochar, there was a rise in the *nosZ* gene, which codes for N_2O reductase, a crucial enzyme for denitrification [354].

To further understand the effects of different levels of biochar on soil nutrient indicators, soil microorganisms and crop growth under the nitrogen-reduction condition, experiment was set up with four treatments as followed: 0%, 2.5% and 5% biochar application rates with 99 kg/hm² nitrogen fertilizer and one control group (the actual fertilizer standard used in the field: 110 kg/hm²) without no exogenous biochar supplement [358]. It was found that a higher application rate of biochar (among 0%, 2.5% and 5% biochar) in soil cultivated with rice led to a significant increased trend in the soil pH, OM and total N content. In addition, a high concentration of biochar under nitrogen-reduction condition decreased the soil bacterial diversity but elevated the fungal diversity. These results provided a theoretical basis for exploring green and efficient fertilization strategies for increasing rice crop yields by appropriate usage of biochar under the nitrogen-reduction background. However, further research on detailed effects of biochar application on rice production is needed for analysis of the physiological functions of the soil microbes and the potential metabolic pathways. In addition, detailed metabolic differences and accurate taxa alteration at the "species" level may be determined using next-generation metagenomic technology.

The impact of biochar application and its combined treatment with PGPB has been studied in numerous field studies and experiments have provided compelling evidence of the positive effects of biochar and PGPB use on plant growth and crop yield in a variety of crop types, from high-value crops like fruits and vegetables to basic cereals like rice and maize [42, 69, 376, 377]. Table 6 summarizes the effect of biochar amendments either singly or along with organic matter or beneficial microorganisms on improving nutrient's availability in soil and its impact of plant growth and crop yield. Research in various agroecosystems has demonstrated that biochar amended soils exhibit increased concentrations of essential nutrients, leading to enhanced plant vigour, root development, and ultimately, higher crop yields [359]. For instance, increased nutrient uptake, biomass and notable increases in yield of maize was favourably correlated with biochar's impact on nutrient availability, suggesting that biochar may increase the yield of staple crops [378, 379]. Biochar addition has also been found beneficial for vegetable crops, including lettuce and tomatoes. Results showed that crops grown in soils treated with biochar had better fruit quality, increased nutritional value, higher commercial yields, and increased disease and insect resistance [380, 381]. Similarly, use of biochar in orchards and tree plantations showed that it increased water retention in the root zone and improved nutrient availability. The improved soil structure facilitated by biochar promoted healthier root development. Increased root biomass and a larger root system are common traits of plants cultivated in soil treated with biochar [382, 383]. Biochar has also been associated with promoting early vegetative growth and flowering in many crops [380]. Increased nutrient availability, better soil structure, and beneficial microbial interactions all contributed to accelerated growth stages, which enabled plants to reach maturity earlier and may lead to higher yields.

The combined treatment of biochar with PGPB further improved the chemical and biological properties of soil and increased plant growth and crop yields. For instance, amendment of biochar prepared from bean husk (BHB) or grape fermentation residue (GFB) with *Bradyrhizobium elkanii* USDA 76 (BRA), *Burkholderia cepacia* ATCC 25416 (PRB), or *Rhizobium altiplani* BR10423 (RHI) improved soil chemical and biological properties in cowpea cultivated in poor soils [394]. BHB and PRB treatments stimulated cowpea growth, whereas GFB and PRB treatments promoted soil phosphatase activity. Different combinations of biochar and plant growth-promoting rhizobacteria (PGPR) increased soil pH, P, K, organic C content, and urease activity. Interestingly, biochars inoculated with the BRA (*B. elkanii* USDA 76) showed highest crop productivity and a 3.85-fold increase in productivity was observed in plants subjected to the BRA + GFB treatment as compared to the use of commercial peat. These findings highlighted the potential of biochar as an environmentally sustainable carrier of PGPB while addressing the issue of agricultural waste reuse. In another study, Chaplot et al. [395] analyzed the experimental data on the impacts of biochar application on crop yield and/or biomass production in agricultural soils. The mean biomass or grain yield gain, which was 16.1% (median at 7.1%) for all available experiments, decreased to -0.64% (median at 5.2%).

Treatments of wheat (*Triticum aestivum* L.) plants with 150 mg L⁻¹ of zinc manganese oxide nanocomposites (ZnO/MnO-NCs) (synthesized from leaf extract of *Conocarpus erectus* L.), 3% w/w corncob biochar, and PGPR strain *Bacillus flexa* enhanced the morpho-physio-biochemical parameters in plants and improved growth of wheat plants, and reduced Cd toxicity [396]. However, among the bio-stimulants tested, PGPR *Bacillus*

Table 6 Effects of biochar fertilization alone or with organic matter/fertilizer, and other beneficial microbes on the nutrient contents in soil, plant growth and crop yield

Treatments	Crop/soil	Physiological effects	References
<i>Biochar treatment</i>			
Wheat straw pellet biochar (WSP) and softwood pellet biochar (SWP) at 2% (w/w)	Maize	Amendment of clay loam soil with biochar in pots enhanced plant biomass and water-use efficiency compared to the unamended controls. The WSP amendment decreased soil net N mineralization rate and it significantly increased the retention of carbon and nitrogen in the soil–plant systems	Wan et al. [51]
Biochar application at concentration of 3.0% (T3.0), and 4.5% (T4.5)	Rice	High application rate of biochar in different treatments significantly improved nutrient availability and soil enzyme activities. Application of biochar at application rates of 1.5% (T1.5), T3.0, and T4.5 increased rice grain yields in comparison to control treatment by 116.38, 141.24 and 145.20%, respectively in a saline-alkali paddy field	Jin et al. [253]
Biochar prepared from tomato straw (TSB), sweet pepper straw (SPSB), and eggplant straw (ESB)	Cucumber	Significantly higher levels of N, P, and K were obtained with SPSB amendment along with superior adsorption capacity as compared to TSB and ESB treatments. SPSB treatment at 5% (w/w) concentration increased plant height by 13.01%, stem thickness by 20.79%, leaf area by 50.26%, and dry weight by 58.56% as compared to the untreated control plants in greenhouse	[63, 64]
Biochar	Soils	The addition of biochar increased the nutrient pool and the turnover of available organic nutrients. Biochar addition altered soil physical and chemical properties and enhanced microbial communities. Biochar addition also influenced nutrient transformations of N, P, K and S, as well as micronutrients including Cu, Fe, Mn and Zn	DeLuca et al. [384]
Biochar	Tobacco (<i>Nicotiana tabacum</i>) varieties LJ911 and LY101	The addition of biochar improved photosynthetic efficiency, enhanced antioxidant capacity, and promoted tobacco growth through hormone synthesis and signaling in both varieties under continuous cropping conditions. Biochar significantly increased the activity of antioxidant enzymes in the leaves of both tobacco varieties under continuous-cropping conditions. Additionally, biochar up-regulated endogenous ABA, auxin, and jasmonic acid (JA) contents in the leaves of both tobacco varieties, with higher levels observed in LY101 than in LJ911 variety	Ye et al. [385]
Biochar amendment (0, 7.5, 15, and 30 t ha ⁻¹) in a sandy loam soil	Black morel (<i>Morchella sextelata</i>)	Higher biochar levels significantly increased fungal α -diversity, promoted saprotrophic, symbiotrophic and ectomycorrhizal guilds, suppressed fungal pathogens, and lowered the absolute abundance of competing fungi. Machine-learning based correlation analysis revealed that the concentration of soil NO ₃ ⁻ -N upon fructification played a key role in microbial community dynamics and morel yield. Biochar amendments altered N availability, reduced fungal competition and increased mushroom productivity	Kang et al. [386]
<i>Biochar and fertilizer treatment</i>			
Biochar (0, 10, 50 and 120 t ha ⁻¹) + fertilization with N, P and K, and micronutrients	Red clover (<i>Trifolium pratense</i> L.)	Biochar addition (at a rate of 10 t ha ⁻¹) increased biological nitrogen fixation (BNF) at the highest rate and showed the greatest increase in total biomass. However, biomass production significantly decreased at an application rate of 120 t ha ⁻¹ in both single and mixed cultures as compared to the control. In the absence of biochar, K fertilization caused a significant increase in BNF. For N, P, and micronutrient fertilization, BNF did not significantly differ between treatments with and without biochar addition	Mia et al. [387]

Table 6 (continued)

Treatments	Crop/soil	Physiological effects	References
Biochar (co-pyrolyzed and mixed with soil) + soluble FePO_4 (Fe-P) or soluble NaH_2PO_4 (Na-P)	Common bean (<i>Phaseolus vulgaris</i>)	Biochar increased arbuscular mycorrhizal fungi (AMF) colonization by 6% and increased Fe-P uptake from 3.1 to 3.8 mg plant ⁻¹ , with AMF-related Fe-P uptake increased by 12%. Biochar promoted the uptake of sparingly soluble P (FePO_4) with AMF hyphae and increased root access to soluble P (NaH_2PO_4) by beans. Soluble P located on biochar increased total plant and microbial P. Biochar reversed reductions in specific root length induced by AMF	Vanek and Lehmann [388]
Biochars derived from maize stalks + slow-releasing phosphorus fertilizers	Soil	Original biochars decreased soil available P through P adsorption and there were no significant differences of soil P fractions under the treatments of mineral P fertilizer and P-laden biochars. On the other hand, P-laden biochars, especially 600°C P-laden biochar, could maintain soil available P at significantly higher level across the incubation compared to mineral P fertilizer to improve soil P use efficiency	[119, 120]
Biochar-based controlled release N fertilizers (BCRNFs) having urea	Oilseed rape (<i>Brassica napus</i> L.)	BCRNF treatment significantly improved soil NO_3^- leading to an increase in N uptake, improved nitrogen-use efficiency (~58.79%), promoted rape growth and thereby enhanced the grain yield (~16.6%) of oilseed rape by slowly releasing N. Treatment modulated the abundance of functional microbes through increased soil nitrification and reduced denitrification, as compared with urea treatment. Bacterial community groups with higher nutrient metabolic cycling ability were found under BCRNF treatment during harvest stage using high-throughput sequencing	Liao et al. [339]
Combined effects of rice straw biochar (0, 10, 20 and 40 t ha ⁻¹), and nitrogen (180, 144 and 108 kg ha ⁻¹ , named N100, N80 and N60)	Rape seed	Application of biochar at a rate of 10 t ha ⁻¹ (B10) replaced 20% N-fertilizer and nitrogen use efficiency (NUE), promoted rapeseed root development and increased rapeseed yield in upland purple calcareous soils. The better management practices were N80B10 and N100B10, which increased NUE by 10.0% and 10.2%, and grain yield by 16.1% and 21.8%, respectively as compared with N100B0	Tian et al. [389]
Plant feedstock biochar and vinegar into the soil (1:2500 g/g; along with chemical fertilizer)	Tomato (<i>Solanum lycopersicum</i> L.)	Significantly improved the yield of tomato plants. Treatment with biochar prepared from chicken manure and animal litter showed higher sugar content, and this increase was attributed to better pH conditions in the rhizosphere soil	Idowu et al. [390]
<i>Biochar and organic matter treatment</i>			
Rice husk biochar (RH; 0%, 2.5%, and 5% conc.) and foliar application of salicylic acid (SA; 0, 0.5, and 1 mM conc.)	Chinese mustard	Treatments increased fresh (19-fold) and dry (three-fold) shoot weight of Chinese mustard. Contents of antioxidant enzymes were significantly improved. Chlorophyll a and b, and carotenoid contents were increased by approximately 300, 400, and 250%, respectively in comparison with the untreated soil. Significant increases in contents of H_2O_2 , MDA, and proline were observed in the HM-contaminated soil. Biochar treatments caused reductions of HMs such as Pb, Cu, Cd, and Zn concentrations upto 74.72, 69.19, 54.16, and 53.86%, respectively, in shoot of Chinese mustard as compared with untreated control plants	Awad et al. [307]
Biochar derived from sewage waste at dose of 20 t ha ⁻¹ and 30 t ha ⁻¹ and coconut husk (CHB_{20} , CHB_{30}),	Maize	All biochar treatments reduced the concentration of NH_4^+ -N by 22–88% in the soil in a greenhouse experiment. CHB_{20} , CHB_{30} and SSB_{20} increased NO_3^- -N by 17–40% in the bulk soil and reduced it by 21–38% in the rhizosphere. Whereas SSB_{30} treatment increased NO_3^- -N by 17–25% in both compartments, but decreased plant N and P uptake efficiency. Biochar increased shoot biomass and P availability by 12% and 12–50%, respectively, except the CHB_{30} , which reduced P availability and N and P uptake efficiency. Biochar improved plant growth during the first 60 days of cultivation in acidic Ultisol soils	Gonzaga et al. [391]

Table 6 (continued)

Treatments	Crop/soil	Physiological effects	References
Exogenous straw addition or biochar-amended soil	Maize	Exogenous straw addition stimulated microbial growth, synthesized a large number of enzymes and accelerated nutrient cycling in maize rhizosphere. The relative abundance of the Firmicutes and the gene copies encoding β -1,4-glucosidase and β -N-acetylglucosaminidase hydrolases were increased. Biochar amendment increased the relative abundance of Actinobacteria in the soil. Biochar enhanced the root biomass by 31%	Shang et al. [65]
Biochar amendment with different oil cakes (viz. castor, coconut, mustard, linseed, and sesame)	Tomato (<i>Solanum lycopersicum</i> L.)	Biochar amendment enhanced plant growth, fresh weight to dry weight, total chlorophyll, carotenoid levels, and nitrate reductase activity. The eggmass/root of the root-knot nematode <i>Meloidogyne incognita</i> , the number of galls and nematode population were reduced with biochar treatments as compared to the untreated control plants	Ikram et al. [267]
<i>Biochar and PGPR treatment</i>			
Biochar and <i>Pseudomonas putida</i> TSAU6	Wheat (<i>Triticum aestivum</i> L.)	The root and shoot dry biomass of wheat plants grown in soil mixed with biochar and combined with <i>P. putida</i> were enhanced significantly compared to the control plants. The N and P contents in plant biomass were increased by biochar addition and biochar combined with a bacterial inoculant. Change in soil enzyme activities, such as higher fluorescein diacetate (FDA) and protease activities were observed after biochar as well as combined application as compared to control soil	Dilfuza et al. [392]
Biochar (BC) + <i>Paraburkholderia phytofirmans</i> s	Soybean	In BC and PGPR-treated plants, maximum increase in photosynthetic (12–30%) and the enzymatic antioxidant activities (5–20) was observed in pots in comparison with control. The combined application with BC and PGPR considerably increased the grain yield (14%) under drought stress conditions	Nawaz et al. [71]
Wood biochar with <i>Bacillus velezensis</i> and an inactivated yeast (IY) biofertilizer	Silver maple (<i>Acer saccharinum</i> L.)	Combined application of biochars with <i>B. velezensis</i> and inactivated yeast significantly increased saplings growth and biomass (91%) as compared to control in greenhouse experiment. Nutrient limitations of both macronutrients (N, P, K, Mg, Ca), and micronutrients (B, Fe, Mn, Mo, Na, S, and Zn) was effectively addressed in combined treatments, and foliar concentrations of Cu was reduced	Sifton et al. [281]
Biochar (BC) and biochar immobilized with the phenanthrene-degrading bacteria (MBC)	Maize	Biochar amendments increased the plant growth and significantly changed the bacterial community structure. Biochar amendment also alleviated the toxicity of PAH-phenanthrene on soil microbes. The BC and MBC treatments showed significantly lower phenanthrene removal rate than the no-biochar treatments	Zhou et al. [316]
Biochar obtained from waste wood of <i>Morus alba</i> solely or in combination with PGPR strains (<i>Pseudomonas</i> sp.)	Canola (<i>Brassica napus</i> L.)	The combined application of both biochar and PGPR increased emergence energy (EE), leaf area index, leaf area ratio (LAR), root shoot ratio (RSR), moisture content of leaves, moisture content of shoot (MCS) and moisture content of root (MCR), and RWC in both varieties of canola under drought stress. The concentrations of N, P, K, Mg, and Ca were highest in all the biochar and PGPRs separate and combined treatments in both varieties, while lowest in 15 and 30 days drought treatments. Osmolyte contents like glycine betaine and sugar remarkably increased in the stress condition and then reduced due to the synergistic application of biochar and PGPR. The antioxidant enzymes and phenolic compounds like POD, SOD, and GR as well as total flavonoids and phenolics were dramatically increased by the combined action of biochar and PGPRs. A significant increase in EE, LAR, RSR, and RWC under 15 and 30 days drought conditions highlighted the synergistic effect of BC and PGPR	Gul-Lalay et al. [393]

Table 6 (continued)

Treatments	Crop/soil	Physiological effects	References
Biochar and arbuscular mycorrhizal fungi (AMF)	Maize	Combined treatment of AMF and biochar significantly increased N uptake in the shoots and roots a pot experiment. Both the – P and +P application rates, significantly increased the root lengths, of 78.43% and 53.09%, respectively, and the root weights of the AMF and biochar combined (AMBC) treatments increased by 58.3% and 43.2%, respectively, compared with the control (CN) treatments. The combined treatment also significantly changed the soil microbe community structure and increased the abundances of <i>Geobacter</i> and <i>Pseudomonas</i> . In addition, the abundances of the N-cycle genes of each process were enhanced. Under the – P condition, the total abundances of the N-cycle genes increased significantly by 1.97–2.19 times in the AMBC treatment compared with the CN treatment	[184, 185]
Biochar + nitrogen-fixing bacteria (<i>Azotobacter chroococcum</i>)	Lettuce	Combined treatment improved lettuce plant nitrogen by 47.39–72.47% and increased root and shoot biomass by 50.35–107.32% through nitrogen fixation and amino acid release. BAC improved the rhizosphere nutrient environment and promoted the deterministic assembly of the rhizosphere microbial community. It also increased microbial diversity and attracted taxa like Actinomycetales (7.59%), Solirubrobacterales (5.17%), Rhizobiales (5.17%), and Sphingomonadales (5.17%), all associated with nitrogen fixation, plant growth promotion and Cd immobilization	Chang et al. [310]
Mixed biochar derived from rice husk and cow bone (MB) + cadmium (Cd)-tolerant plant growth-promoting bacteria (MBM)	Radish	Treatment with MBM enriched biochar significantly increased the abundance of phosphorous cycling functional genes <i>pqqC</i> by 299.98% and <i>phoD</i> by 384.93% respectively. Moreover, MBM significantly increased soil available P content by 340.09%, the activity of radish peroxidase, superoxide dismutase and catalase were increased by 162.67, 21.52 and 65.10%, respectively and radish shoot/root biomass by 141.53% and 92.69%. MBM significantly enhanced plant growth and soil health through a synergistic interaction between the biochar and microbial inoculants	Gong et al. [309]
Rice biochar (RBC) + <i>Metabacillus indicus</i> AMB4 (PGPR) treatment	Rice	Combined RBC + PGPR treatment enhanced soil nutrient status and yielded 56% more tillers and 73% more panicles in rice as compared to control plants under field trials. Combined treatment also reduced soil arsenic bioavailability by 85% and plant arsenic accumulation by 78% compared to controls, and significantly enhanced antioxidant enzyme activities (CAT: 4.2-fold, SOD: 3.7-fold, APX: 3.1-fold increase). Combined treatment also restructured microbial communities with increased Proteobacteria abundance and upregulated arsenic detoxification genes (<i>arsC</i> , <i>arsB</i> , <i>aioA</i>) coupled with enhanced nutrient cycling pathways (<i>nifH</i> , <i>phoD</i> , <i>soxA</i>)	Majumdar et al. [73]
Biochar prepared from coconut coir (CC), coconut shell (CS), rice husk (RH), and sugarcane bagasse (SB) + <i>Bradyrhizobium</i> strains KNUST 1002 and KNUST 1006 and reference strain BR 3267	Cowpea (<i>Vigna unguiculata</i> L.)	<i>Bradyrhizobium</i> strain 3267 in CC and SB significantly increased the nodule dry weight of cowpea as compared to uninoculated control under field conditions. The highest grain yield (1774 kg ha ⁻¹) was recorded for CC, which was 16% greater than the grain yield of the peat-based inoculant (1524 kg ha ⁻¹). Treatment with CC and SB enhanced <i>Bradyrhizobium</i> survival for up to 16 and 20 weeks, respectively, whereas CS and RH promoted strain survival for up to 8 weeks. The viable cell count was highest in the peat-based inoculant during the 24-week period	Mensah et al. [50]

Table 6 (continued)

Treatments	Crop/soil	Physiological effects	References
Biochar prepared from moss + halo-tolerant <i>Halomonas salifodinae</i>	<i>Medicago sativa</i> grass	Biochar remarkably assisted the root binding and biofilm formation of halotolerant <i>Halomonas salifodinae</i> on the <i>Medicago sativa</i> grass roots. The combined treatment of biochar and bacterium increased the numbers of total bacteria, halo-tolerant bacteria and nitrogen-fixing bacteria. The combined treatment also increased the plant height from 10.3 cm to 36 cm, and enhanced the accumulation of chlorophyll a, reducing sugars, soluble proteins, and superoxide dismutase in the leaves after 30 days of culturing	[66–68]
Biochar and PGPR	Tomato	Combined treatment increased tomato seedling biomass as compared to the control. However, bacterial diversity was higher only in sole PGPR treatment. Application methods also differentially affected the soil bacterial community composition and community structure. In addition, the application of biochar alone or with PGPR increased soil pH and C:N ratio, and improved soil health and crop plant growth as compared to control	Yan et al. [72]

flexa sp. was most effective in lowering Cd accumulation in the roots and shoots, showing a 15.35% and 17.71% increase in root length and 5.28% and 7.02% increase in shoot length compared to nanocomposites and biochar, respectively in the presence of Cd stress. It also enhanced chlorophyll content and antioxidant enzyme activities while reducing oxidative stress markers. The soil-drenching application of these bio-stimulants showed higher efficiency to improve wheat growth, crop yield and reduction in uptake of Cd stress. Metagenomic and metabolomic analyses were used to evaluate three biochar application strategies [66–68] and results showed that furrow-applied biochar (BR4) treatment at a reduced rate (1,875 kg ha⁻¹) enhanced soil microbial structure, function, and metabolite composition, and ultimately significantly increased sesame yield by 26.57% as compared to no biochar control treatment (BR0) in continuous sesame cropping systems. The results showed that BR4 biochar treatment and surface-applied BR3 biochar treatment (@15,000 kg ha⁻¹) enhanced the relative abundance of Actinomycetota while suppressing *Pseudomonas* in rhizosphere and non-rhizosphere soils of diseased plants. *Sphingobium amiense* and *S. indicum* were differentially abundant in BR3 and BR4 amended healthy rhizosphere soil, respectively. Biochar application significantly reduced bacterial wilt and boosts yield in continuous sesame cropping.

In a recent study Silva et al. [376, 397] inoculated five diazotrophic isolates on *Brachiaria brizantha* forage systems, with biochar (as a pelletizing material) or without biochar, and plant traits and soil attributes were evaluated. All isolates increased soil K content, with the highest 39% increase in the biochar amended treatments. Soil pH and urease activity were significantly modulated by different isolates, while biomass allocation patterns differed among treatments. Overall, the results highlighted that biochar pelletization can enhance the effectiveness of diazotrophic bacteria. Machine learning (ML) revealed that dry foliar biomass, soil pH, and fresh root weight were the most predictive variables, highlighting consistent signatures explaining plant–soil responses to biochar-pelletized diazotrophic bacteria. In another experiment, Xu et al. [69] effectively immobilized the drought-tolerant PGPR (*Bacillus thuringiensis* IP3), in coconut-shell biochar and applied to masson pine (*Pinus massoniana*) conifer. The combined application of biochar and PGPR significantly increased the seedling relative water content by 40.1% relative to the control under drought stress. The co-application of biochar–PGPR

agent enhanced both shoot and root growth, increased the net photosynthetic rate 7.4-fold, and raised POD activity by 74.2%, CAT activity by 66.3%, and proline content by 40.3% under drought stress. In addition, the biochar–PGPR treatment significantly improved soil nutrients, altered rhizosphere microbial composition and function, stimulated soil amino-acid and lipid metabolic pathways, and increased the relative abundances of the nutrient-cycling and biofilm-related genes.

Although biochar amendment clearly improves nutrient availability, enhances biomass production, stimulates plant growth and increases crop yields in agricultural soils, a number of factors need to be taken into account to optimize its benefits and guarantee sustainable farming methods.

- i. Feedstock variation: The nutritional content and efficacy of biochar is affected by the type of biomass utilized in its formation. Understanding the differences between the nutritional profiles of biochar made from different feedstocks is essential for developing targeted soil amendments for particular crops and soil types [398].
- ii. Application rates and timing: The optimal application rate of biochar depends on factors such as soil type, climate, and crop requirements. By using biochar at the appropriate growth stage in the crop cycle and nutrient release alignment with the plant's needs, avoid excesses or deficiencies of nutrients [377].
- iii. Interactions with other soil amendments: Biochar is frequently combined with other soil amendments, including mineral nutrients or organic fertilizers for enhancing plant growth and crop yield. Designing of successful nutrient management strategies requires an understanding of the synergies or possible conflicts between biochar and these amendments [399].

7 Constraints in development, application and marketability of biochar

The development, application, and marketability of biochar, despite its promising attributes, face several constraints that hinder its widespread adoption and integration into mainstream agricultural practices. These constraints span technological, economic, and social dimensions, requiring careful consideration for the effective utilization of biochar in addressing environmental and agricultural challenges [400]. One of the primary challenges associated with use of biochar is the cost of production. The pyrolysis process, which transforms biomass into biochar, requires energy inputs and specialized equipment, contributing to production costs. Scaling up production to meet the demands of large agricultural areas can be economically challenging. As a result, the cost-effectiveness of biochar production needs improvement to ensure its accessibility to a broader range of farmers [400, 401].

Availability and variability of feedstock also pose challenges in maintaining consistent biochar quality as the type of biomass used as feedstock for biochar production significantly influences its properties. Different feedstocks yield biochars with distinct physical and chemical characteristics, impacting their effectiveness in soil amendment [402]. Addressing this challenge requires research into optimizing biochar production from diverse feedstocks while ensuring reliability and uniformity. In addition, the absence of standardized protocols and certification procedures for biochar products hampers marketability and consumer confidence. The lack of clear quality standards makes it difficult for farmers to choose biochar products that meet specific needs [403]. Establishing

industry-wide standards and certification processes is crucial for fostering trust among users and facilitating the adoption of biochar on a larger scale [75].

Although laboratory and greenhouse studies show encouraging results, applying biochar at the field scale is not always straightforward. Its performance can vary widely depending on differences in soil conditions, climate conditions and crop management practices. Applying biochar at the field scale is influenced by various practical and environmental factors such as soil variability, as soil properties vary across a field leading to uneven crop responses after biochar application. This makes it important to adopt site-specific strategies that are according to local soil conditions [404, 405]. In addition, the cost of producing and transporting biochar in large quantities can also be a major limitation, especially for farmers with limited resources. Using locally available feedstocks offer a more practical and cost-effective solution [406, 407]. Another key limitation is the way biochar is applied in the field. If it is not evenly distributed, the benefits can vary across the field, leading to inconsistent results. Thus, mechanized application can help achieve more uniform results [408]. There is also limited information on how biochar behaves over the long-term under field conditions, particularly in terms of nutrient cycling in soil, microbial activity and possible accumulation effects. To improve its practical use, biochar should be integrated with existing soil management practices, such as composting, integrated nutrient management and conservation agriculture [409, 410].

Another major constraint in the application of biochar is the risk of its contamination with HMs (e.g., Cd, Pb, Zn), which is particularly relevant when biochar is produced from wastes containing such pollutants such as sewage sludge, swine manure or industrial residues [411, 412]. Also, incomplete pyrolysis process can result in the formation of toxic compounds like polycyclic aromatic hydrocarbons [413] and VOCs, which may affect soil biota and plants. Other pollutants such as environmental persistent free radicals (EPFRs) present with biochar can induce production of reactive oxygen species, which can destabilize cell membranes and induce apoptosis resulting in repressed growth [414]. Besides all these factors, some studies indicate that weathering and aging of biochar can lead to formation of nanobiochar [411, 415], which can transport pollutants with it more quickly in environment. Biochar is often regarded as a cost-effective option for soil remediation. However, its ability to act both as a sink and a potential source of pollutants forces us to limit its use.

The lack of awareness among farmers and landowners about the potential benefits of biochar is another major constraint in agriculture-based countries. Further, the use of biochar is still relatively novel, and farmers may be hesitant to adopt a technology that involves altering traditional agricultural practices. Concerns about potential side effects, uncertainties about long-term impacts, and the perceived risks associated with biochar use can deter farmers from incorporating it into their farming systems [416, 417]. Adequate training and outreach programs are needed to raise awareness about biochar's efficacy in improving soil fertility, enhancing plant growth, and mitigating environmental issues [416]. Further, the integration of biochar into conventional farming practices may face resistance due to established norms and routines. Demonstrating the compatibility of biochar with existing agricultural systems and highlighting its ease of application is essential for overcoming barriers to its adoption. Bridging this knowledge gap is essential for encouraging widespread adoption of biochar application in the crops [418].

Besides this, there is a need for more crop-specific research to optimize its application for different agricultural systems. Variability in soil types, climates, and crop requirements necessitates targeted investigations to determine the most effective biochar application rates, timing, and integration strategies for various crops. Building the necessary infrastructure, including pyrolysis facilities and distribution networks, is crucial for making biochar readily available to farmers. Investments in infrastructure development can contribute to overcoming this constraint. Overcoming these challenges requires a multidimensional approach involving technological innovation, education and outreach, policy support, and collaborative efforts among researchers, farmers and industry stakeholders. Addressing these constraints is essential to unlock the full potential of biochar as a sustainable and effective solution in the realm of agriculture and environmental management.

8 Conclusions and future perspectives

This study highlights the significant potential of biochar as a sustainable alternative for improving soil physical properties, enhancing carbon sequestration and mitigating greenhouse gases. The article clearly describes that the effectiveness of biochar is strongly influenced by feedstock type, pyrolysis conditions and soil properties. Also, the role of biochar in stabilizing carbon through physicochemical protection and microbial interactions, demonstrates its importance in mitigating climate changes.

The future perspectives of biochar hold immense promise to address pressing challenges in agriculture, environmental sustainability and climate change mitigation. Biochar's role in carbon sequestration presents a significant opportunity to mitigate climate change. By converting biomass into a stable form of C through pyrolysis, biochar locks away carbon in the soil, reducing atmospheric CO₂ levels. As governments and industries increasingly focus on carbon-neutral practices, biochar stands out as a valuable tool in the arsenal of climate change mitigation strategies. Other than this, its ability to improve soil structure, retaining nutrients, and stimulation of microbial activity contributes to sustainable agriculture. As global demand for food continues to rise, incorporating biochar into farming practices can lead to increased crop yields, improved soil resilience, and reduced dependence on chemical fertilizers [381].

Biochar production provides an avenue for recycling organic waste materials, turning them into a valuable resource. Agricultural residues, forestry by-products, and organic waste can be converted into biochar, reducing the environmental impact of waste disposal. This aligns with the principles of a circular economy, where waste is transformed into a resource, contributing to a more sustainable and environmentally friendly approach to waste management. Besides this, biochar's ability to adsorb pollutants can be exploited for water pollution treatment also. Application in water treatment can help improve water quality by capturing and removing contaminants. This makes biochar a potential solution for addressing water pollution in agricultural runoff and mitigating the impacts of industrial activities on water resources.

Despite these promising outcomes, several challenges remain. Technologically, there is a need to optimize pyrolysis processes to ensure consistent biochar quality while keeping in view the issues related to feedstock variability and large-scale production. Ongoing research and technological advancements are likely to refine biochar production processes, enhancing efficiency and reducing costs [402]. Innovations in feedstock

selection, pyrolysis techniques, and the development of designer biochars with specific properties for targeted applications will further optimize the efficacy of biochar in diverse contexts. The biochar market has many different opportunities, which can reduce risks and expand its overall potential. However, this diversity can also make it harder for new businesses to stay focused, especially in the early stages. Small but profitable niche markets—such as horticulture, nurseries, water treatment, and energy-related uses—are often the best starting points. For entrepreneurs with limited resources, focusing on one specific market at the beginning is important to find the right product-market fit. Even after choosing a target market, there are still challenges and barriers that can slow growth or affect business success. Therefore, it is essential for biochar producers to identify these challenges early and plan effective strategies to overcome these problems [419]. In conclusion, the future perspectives of biochar are bright, with the potential to revolutionize agriculture, its contribution to climate change mitigation, and in addressing waste management challenges. As research continues to uncover its diverse applications and benefits, the integration of biochar into mainstream practices holds the key to a more sustainable and resilient future for agriculture and the environment.

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Author contributions

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